

Stochastic Choice from Gradual Attention

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Abstract

We propose a novel model of stochastic choices with limited attention. A consumer evaluates a menu of options sequentially, following an order that depends on the salience of the options (called *salience order*). Due to the time constraint, the consumer might not be able to evaluate every option in the menu. The set of options she has evaluated forms her consideration set and she chooses her most preferred option from the consideration set. Her evaluation process is modeled by a counting process. As a consequence, her consideration set and choice can be stochastic. We provide an axiomatic characterization of this model. With the rich data containing both the choices and the associated contemplation time, we show the primitives of the models such as the salience order and the preference can be uniquely recovered. A key feature of our model is that the consumer makes better choices (in terms of first-order stochastic dominance) as her contemplation time increases. We test this choice pattern using the experimental data by Caplin and Dean (2011), and find suggestive evidence in support of this choice pattern.

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1 Introduction

The classic revealed preference theory assumes a consumer considers all available options and selects the best one according to a preference relation. A substantial empirical studies found evidence against the classic revealed preference theory. A growing literature shows some of these choice anomalies can be attributed to the limited attention, which assumes a consumer makes choices from a subset of all available options, known as the consideration set. The idea of limited attention is also practically motivated by the fact that in reality there are often too many options (e.g., different brands of juices in the supermarket) presented to consumers so they might ignore some of the available options. Early works (Masatlioglu, Nakajima, and Ozbay (2012), Lleras et al. (2017), and Dean, Kıbrıs, and Masatlioglu (2017)) in limited attention models assume the consideration set to be deterministic and so is the choice behavior. A few recent works (Manzini and Mariotti (2014) and Brady and Rehbeck (2016)) have extended the deterministic limited attention model to allow for stochastic attention, in which the consumer pays attention to a collection of subsets, governed by a probability distribution. As a consequence, choices can be stochastic. Allowing stochastic choices adds more flexibility to the models’ empirical applicability.

Most existing models in stochastic choice with limited attention parameterize the formation of consideration sets in specific ways without providing any behavioral justifications. In the deterministic choice settings, Masatlioglu and Nakajima (2013), Caplin, Dean, and Martin (2011), and Caplin and Dean (2011) model the formation of a consideration set as a consequence of searching. In this paper, we also consider a search-based consideration set formation process, while allowing the consideration set to be stochastic. We argue this searching-based consideration formation is appealing since studies (Reutskaja et al. (2011)) using eye-tracking data shows a consumer is indeed evaluating items sequentially, instead of simultaneously paying attention to all items.

In particular, we assume the items can be ranked by their salience and a consumer evaluates each item sequentially, following this salience order. For example, this salience order can be the list order in the context of online shopping at Amazon.¹ The size of the consideration set evolves according to a counting process, e.g., Poisson process. The counting process can be viewed as a way to model the consumer’s evaluation speed. Given the salience order, the consideration set is growing monotonically over time. When search stops, the consumer might not be able to evaluate all items so her consideration set might be a strict subset of all available items. She then chooses the best item from this consideration

¹See Rubinstein and Salant (2006), Rubinstein and Salant (2011), and Yildiz (2016) for studies that assumes consumers consider items following certain orders.

set, according to her preference.

The monotone growing (stochastic) consideration set leads to the key feature of our model, i.e., a consumer must make better choices as the contemplation time increases. We call this choice pattern “increasing accuracy”.² Note this feature relies on the implicit assumption that the time constraint is predetermined (or exogenous).³ In traditional search models, due to the searching/time cost, a consumer terminates searching following an optimal stopping rule. This optimal stopping rule can depend on the current consideration set, e.g., searching stops if the best item in the current consideration set is sufficiently valuable, as the satisficing searching model proposed by Simon (1955). It is therefore not surprising that the choice pattern of better choices over time predicted by our model is opposite to search models with an optimal stopping rule such as Fudenberg, Strack, and Strzalecki (2018), where they demonstrated a negative correlation between contemplation time and accuracy. In Section 6, we revisit the experiment by Caplin and Dean (2011) and use their data⁴ to test the increasing accuracy implication of our model. We found suggestive evidence in support of this choice pattern. We believe whether the contemplation time is endogenous or exogenous depends on the application. But an endogenous optimal stopping rule certainly requires a consumer to have more knowledge about the searching, such as the distribution of values of the items, while our modeling approach is “prior free”.

We provide an axiomatic characterization of the model in Section 2. Empirically testing these axioms requires richer data than standard stochastic choice models. We assume researchers not only observe choices from menus, but also observe the amount of time that is used to make a decision. For example, the contemplation time can be measured by the amount of time a consumer used to find a restaurant in Yelp. We show the primitives such as the preference and the salience order can be uniquely recovered from the richer data. Intuitively, an item a is revealed to be preferred to b if it is always chosen against b or its choice probability from the doubleton menu $\{a, b\}$ is increasing over time. Similarly, a is revealed to be more salient than b if it is always chosen or its choice probability is decreasing over time because the consumer considers a ahead of b .

The static version of our model involves no choice variations over time. We show in general, the preference and the salience order can only be partially identified in the static model. This negative result highlights the novelty of making use of choice variations over time as exact identification of the primitives can be achieved. The static model can be

²This feature is also shared by the other search-based limited attention models, e.g., Masatlioglu and Nakajima (2013), Caplin, Dean, and Martin (2011), and Caplin and Dean (2011).

³Caplin, Dean, and Martin (2011) and Caplin and Dean (2011) intentionally design their experiment in a way such that the consumer cannot influence their stopping time.

⁴To be precise, their data contain more details than required by our model.

directly compared with the existing limited attention models and it turns out to be a special case of the monotone random attention model by Cattaneo et al. (2017). Thus all their results can be applied directly to our static model.

So far our discussion assumes the stochastic choice data is generated by a consumer making repeated choices. It is common to observe stochastic choices as the aggregated choices from the population, e.g., the market shares of vehicles. Suppose there is population heterogeneity in the primitives (i.e., preferences, salience orders and counting processes) and each individual in the population behaves according to our model. Does there exist a representative agent that behaves according to the model if we only observe their aggregated choices?

⁵ It turns out a representative agent exists under heterogeneity in counting processes. A representative agent might not exist if we allow heterogeneity in preferences and/or salience orders. We characterize the necessary and sufficient conditions for a representative agent to exist when the heterogeneity occurs in preferences and/or salience orders. When there is no cross-menu variations in the data (i.e., we observe only the choices over time for a single menu), we show a representative agent always exists. Therefore the increasing accuracy implication within a menu is robust to any heterogeneity, which is crucial for our empirical analysis using the experimental data by Caplin, Dean, and Martin (2011).

⁵The study on the existence of a representative agent in the context of stochastic choice is scarce. Apesteguia and Ballester (2016) shows a representative agent exists for the general random utility model but does not exist for its widely used special cases such as the Luce model. We are not aware of any similar study for the limited attention models.

2 Gradual Attention Model

2.1 Notations and Assumptions

Let X be a finite set of items. A menu A is a non-empty subset of X and we use $\mathcal{X} = 2^X \setminus \emptyset$ to denote the set of menus. Let $t \in \mathcal{T}$ be the amount of time used for the decision maker (DM) to make a decision, where $\mathcal{T}$ is the time index set. A random choice rule with time $\rho : X \times \mathcal{X} \times \mathcal{T} \rightarrow [0, 1]$ satisfies $\sum_{a \in A} \rho(a, A, t) = 1$ for all $t \in \mathcal{T}$. The amount of time t can be interpreted as the deadline for the decision making problem and hence $\rho(a, A, t)$ is the probability that a being chosen the menu A , given the deadline t . This deadline is viewed as part of the decision problem and is hence exogenous. The time index $\mathcal{T}$ can be discrete (e.g., number of minutes) or continuous. Given a menu A , we denote the support of A as the set of items that is chosen with strictly positive possibility at some time $s \in \mathcal{T}$, i.e., $\text{supp}(A) := \{a \in A : \rho(a, A, s) > 0 \text{ for some } s \in \mathcal{T}\}$.

Let $\mathcal{S}$ denote the set of strict preferences over X , i.e., the set of binary relations that are complete, transitive and anti-symmetric. We use $\succ \in \mathcal{S}$ to denote a preference. Given a preference $\succ \in \mathcal{S}$, the $\succ$ induced choice function, $\mathcal{C}_\succ : \mathcal{X} \rightarrow X$ is defined as $\mathcal{C}_\succ(A) = \{a \in A : a \succ b \text{ for } \forall b \in A\}$. The choice function $\mathcal{C}_\succ$ chooses the best item from a menu, according to the preference $\succ$.

Let $\{N_t\}_{t \in \mathcal{T}}$ (or N for convenience) be a counting process and we will use the realization of the counting process to model the size of the consideration set. For any time $t \in \mathcal{T}$, we denote μ_t^N as its induced probability distribution over $\mathbb{Z}^+$, i.e., $\mu_t^N(Z) := \Pr(N_t = Z)$. Throughout the paper, we restrict our focus on the counting processes that have the following two properties: 1) $\mu_t^N(0) = 0$ for any $t \in \mathcal{T}$ ⁶ and 2) for each $1 \leq Z \leq |X|$, there exists some $t \in \mathcal{T}$ such that $\mu_t^N(Z) > 0$. The second property is satisfied by most commonly used counting processes such as the Poisson process. The set of the counting processes that satisfy these two properties will be denoted by $\mathcal{N}$.

We use $R \in \mathcal{S}$ to denote a “salience order” and aRb implies that item a enters the consideration set earlier than b . Given $Z \in \mathbb{Z}^{++}$, a menu $A \in \mathcal{X}$ and a salience order $R \in \mathcal{S}$, the R induced Z -consideration set of menu A , $A^R(Z)$, is defined as $A^R(Z) := \{a \in A : |\{b \in A : bRa\}| < \min\{Z, |A|\}\}$. In other words, $A^R(Z)$ contains the most salient Z items of the menu A , if $Z < |A|$. If $Z \geq |A|$, then $A^R(Z) = A$.

Definition 1. (Gradual Attention Representation)

A random choice rule with time ρ is said to have a gradual attention representation $(\succ, R, N)$

⁶The purpose of this restriction is to guarantee that the consideration set is non-empty. Otherwise, we need to allow the DM to choose a default/outside option when the consideration set is empty. For any counting process without this property, we can easily modify it by uniformly adding 1 to the realization.

with the preference $\succ$, the salience order R and the counting process N , if for all $a \in X, A \in \mathcal{X}, t \in \mathcal{T}$,

$$\rho(a, A, t) = \sum_{\{Z \in \mathbb{Z}^{++} : \mathcal{C}_{\succ}(A^R(Z))=a\}} \mu_t^N(Z) \quad (1)$$

The gradual attention representation describes the following decision making process: The DM has a true preference $\succ$ and is equipped with a salience order R . She needs to make a choice from a menu A by the deadline t . She may not be able to pay attention to all the items in A due to the time constraint. The items enter her consideration stochastically, following the salience order R . The size of the consideration set depends on the the realization of the counting process N at time t . She then chooses the best item from the realized consideration set according to her true preference $\succ$.

The formation of the consideration set can be alternatively interpreted as learning/searching (at no cost) by a given deadline. The DM assesses the items following from the most salient one to the least salient one. It takes time for the DM to evaluate an item before it enters her consideration set and the amount of time used for evaluation is stochastic. This interpretation is in contrast to the literature of the rational inattention, where the DM optimally chooses a time point to stop learning/searching due to the cost.

In this section, we will discuss identification of the primitives $(\succ, R, N)$ in the gradual attention representation [Equation 1](#) and then give an axiomatic characterization. Unlike the standard stochastic choice literature which exclusively relies on choice variations across menus, the identification in our model also relies on choice variations over time. To guarantee that the choice variations over time are sufficient for identification, we introduce the following time variation assumption.

Assumption A. (Time Variation)

A random choice rule with time ρ is said to satisfy the time variation assumption, if for any menu $A \in \mathcal{X}, a \in A$, $|\rho(a, A, \mathcal{T})| = 1 \iff \rho(a, A, \mathcal{T}) \subset \{0, 1\}$, where $\rho(a, A, \mathcal{T}) = \{\rho(a, A, t) : t \in \mathcal{T}\}$.

The time variation assumption requires that the choice probability of any item to vary over time, unless this item is always chosen or never chosen. One extreme case that fails this assumption is that there is no variation over time at all, which reduces our model to a static version. We will discuss the static version of the model and its identification issues in more details in the section of lack of time variation.

2.2 Identification and Monotonic Feature

We first discuss the monotonic feature of the gradual attention model, which leads to simple identification of the primitives. It turns out the identification of $\succ, R$ can be achieved via the choice variations over time in doubleton menus. For any $t \in \mathcal{T}$, let $\Delta_{ab}(t) := \rho(a, \{a, b\}, t) - \rho(b, \{a, b\}, t)$. We say a dominates b if $\Delta_{ab}(t) = 1$ (or equivalently $\Delta_{ba}(t) = -1$) for all $t \in \mathcal{T}$. We say a impacts b if $\Delta_{ab}(t)$ is weakly increasing in t but is not constant, denoted by $\Delta_{ab}(t) \uparrow$ (or equivalently $\Delta_{ba}(t) \downarrow$).

Suppose ρ has a gradual attention representation [Equation 1](#) and consider the dynamics of choice probabilities in the menu $\{a, b\}$. If $a \succ b$, then the only way b could be chosen is when the consideration set only contains b . This requires b to enter the consideration set earlier than a , which implies bRa . As time increases, the probability that the consideration set being singleton will decrease and so does the choice probability of b .⁷ This reasoning can be summarized as the following proposition.

Proposition 1. (*Identification from Doubletons*)

Suppose ρ satisfies the time variation assumption and has a gradual attention representation $(\succ, R, N)$. For any $a, b \in X$, $\Delta_{ab}(t)$ is monotonic in t and is constant if and only if $\Delta_{ab}(t) = 1$ or -1 . Moreover,

- a dominates b , i.e., $\Delta_{ab}(t) = 1 \iff a \succ b, aRb$
- b dominates a , i.e., $\Delta_{ab}(t) = -1 \iff b \succ a, bRa$
- a impacts b , i.e., $\Delta_{ab}(t) \uparrow \iff a \succ b, bRa$
- b impacts a , i.e., $\Delta_{ab}(t) \downarrow \iff b \succ a, aRb$

It follows that when $|X| = 2$, ρ has a gradual attention representation with a unique pair of $\succ, R$ if and only if $\Delta_{ab}(t)$ is monotone.

The Proposition 1 shows how to identify the $\succ, R$ over an arbitrary pair of items by observing the choice dynamics from their doubleton menu. Since $\succ, R$ both belong to $\mathcal{S}$ and thus are complete, we can then recover $\succ, R$ by checking each doubleton menu. When $|X| = 2$, the monotonicity of $\Delta(t)$ characterizes the representation. With more items in the menu, we need to generalize the monotone $\Delta(t)$ to monotone movement of the entire choice probability distribution over time. We introduce the following notion of the first order stochastic dominance to compare two probability distributions.

⁷One concern for this reasoning is that it is possible that both a, b are being chosen with strictly positive probability but their choice probabilities stay constant over time. Fortunately this possibility is ruled out by the time variation assumption.

Definition 2. (First Order Stochastic Dominance (FOSD))

Let A be a set that is ordered by $P \in \mathcal{S}$. Let ω, ν be two probability distribution over A . We say ν first order stochastically dominates (FOSD) ω with respect to P if for any $a \in A$,

$$\sum_{\{b \in A: aPb\}} \omega(b) \leq \sum_{\{b \in \mathcal{P}: aPb\}} \nu(b)$$

We denote $\omega \succ_F \nu$ w.r.t $>_P$.

Proposition 2. (*Increasing Accuracy over Time*)

Suppose ρ has a gradual attention representation $(\succ, R, N)$. Then for any menu A , $\rho(\cdot, A, t) \succ_F \rho(\cdot, A, s)$ w.r.t $\succ$ for all $t \geq s$.

It is a well known result that $\omega \succ_F \nu$ w.r.t a preference order $\succ$ if and only if ω is preferred to ν by any DM who is maximizing the expected utility using the same preference as $\succ$. It follows that the Proposition 2 can be interpreted as the DM will make better choices in terms of higher expected utility over time.

It is natural to expect that as the amount of time used for decision making gets sufficiently large, the DM should choose the $\succ$ -best item almost surely. The following corollary confirms this intuition, under a mild assumption on the counting process that guarantees that the consideration set eventually converges to the whole menu. This assumption is also satisfied by most commonly used counting processes such as the Poisson process.

Corollary 1. (*Gradual Rational Choice*)

Assume for any $\epsilon > 0$, $K \in \mathbb{Z}^{++}$, there exists $s \in \mathcal{T}$ such that $Pr(N_s < K) =: \mu_s(< K) < \epsilon$. Then ρ converges to $\mathcal{C}_\succ$ in probability as $t \rightarrow \infty$, i.e., for any menu A , $\rho(\mathcal{C}_\succ(A), A, t)$ converges to 1.

The above Corollary 1 shows that the classic deterministic rational choice model is a limiting case of our model. The Proposition 2 tells us this convergence is monotonic in the sense that the expected utility loss due to the deviations from the rational choice monotonically vanishes over time.

2.3 Axiomatic Characterization

We discuss axiomatic characterizations of the gradual attention model. For any $a \in X$, we denote the set of items that impacts a or dominates a as $\overline{M}(a) := \{b \in X : \Delta_{ab}(t) \downarrow \text{ or } \Delta_{ab}(t) = 0\}$. We denote the set of items that are impacted or dominated by a as $\overline{H}(a) := \{b \in X : \Delta_{ab}(t) \uparrow \text{ or } \Delta_{ab}(t) = 1\}$.

Axiom 1. (Monotone Choices from Doubleton Menus)

For any $a, b \in X$, $\Delta_{ab}(t)$ is monotone in t .

Axiom 2. (Dominance Transitivity)

If a dominates b and b dominates c , then a dominates c .

Axiom 3. (Impact Transitivity)

If a impacts b and b impacts c , then a impacts c .

The Axiom 1 says no choice reversals over time is allowed for doubleton menus. So we can compare a generic pair of items in terms of their dominance or impact relation by the Proposition 1. In particular, the proposition then implies $\overline{M}(a)$ coincides with the set of items that are revealed to be preferred to a . From the normative perspective, a dominating or impacting b both suggests a is better than b since the choice probability of a is increasing over time. Under the Axiom 1, the transitivity across impact and dominance is implied by the Axiom 2 and 3 as the following lemma.

Lemma 1. (*Transitivity across Impact and Dominance*)

Assume Axiom 1-3. If a impacts b and b dominates c , then a impacts c or a dominates c ; if a dominates b and b impacts c , then a impacts c or a dominates c .

So far the Axioms 1-3 impose restrictions on ρ only for doubleton menus. The remaining axioms build connections between ρ on doubleton menus and more general menus. The Axiom 4 extends the notion of the dominance to arbitrary menus, with the interpretation that if b dominates a , then a will never be chosen in any menu containing both a, b . Notably, the Axiom 4 alone is also implied by the classical deterministic choice model as well as the random utility models. The Axiom 5 generalizes the Axiom 1 to arbitrary menus, requiring the movement of choice probabilities over time to be monotonic in terms of FOSD w.r.t the revealed preference.

Axiom 4. (Independence of Dominance)

For any menu A , $\rho(a, A, t) = 0$ for all $t \in \mathcal{T}$ if and only if there exists $b \in A$ such that b dominates a .

Axiom 5. (Increasing Accuracy over Time)

Let menu A and $t \geq s$ be arbitrary. For any $a \in A$, $\sum_{c \in \overline{M}(a) \cap A} \rho(c, A, t) \geq \sum_{c \in \overline{M}(a) \cap A} \rho(c, A, s)$.

Theorem 1. (*Gradual Attention Representation*)

Under the assumption of time variation, ρ has a gradual attention representation $(\succ, R, N)$, where $\succ, R$ are menu independent and unique, if and only if ρ satisfies Axioms 1-5.

The only if part of the theorem follows immediately from the Propositions 1, 2. We give a sketch of the proof here and a formal proof is in the appendix. The Axiom 1 allows us to construct a candidate pair of $\succ, R$ from ρ on doubleton menus by the identification results in the Proposition 1. The Axiom 2,3 are then used to prove that the constructed $\succ, R \in \mathcal{S}$. The constructed $\succ, R$ will impose some restrictions on the support of ρ on an arbitrary menu, i.e., an item a is never being chosen if and only if there exists another item b such that $b \succ a, bRa$. This support compatibility condition is satisfied by the Axiom 4. Given $\succ, R$, we can convert the choice probabilities into probabilities over $\mathbb{Z}^{++}$. The Axiom 5 then implies the converted probabilities over $\mathbb{Z}^{++}$ can be ordered by FOSD w.r.t the natural numbers over time. By the Lemma 6, we can then construct a counting process N that induces the probabilities over $\mathbb{Z}^{++}$. The menu independence of $\succ, R$ comes from this construction and the uniqueness follows immediately by the Proposition 1.

The menu independence and the uniqueness of the counting process N are not obtained in the Theorem 1. It can be easily checked by an example in the appendix that two different counting processes can induce the same probability distribution μ over $\mathbb{Z}^{++}$. Since ρ only depends on μ in the representation, we will focus on the induced μ . It is straightforward to see μ can be uniquely identified at most to $|X|$. We first show that if ρ has a gradual attention representation with a menu independent induced μ , then μ is uniquely identified up to an integer that is generally less than $|X|$. It follows that full identification occurs if and only if every menu has full support.

Proposition 3. (*Identification of μ*)

Suppose ρ has a gradual representation with menu-independent $\succ, R, N$, then the μ induced by N is unique up to $\max_{A \in \mathcal{X}, a \in \text{supp}(A)} |\overline{H}(a) \cap A|$. In particular, $\max_{A \in \mathcal{X}, a \in \text{supp}(A)} |\overline{H}(a) \cap A| = |X| - 1$ if and only if $\text{supp}(B) = B$ for any menu B .

The Axiom 6 is added to characterize the gradual attention representation with a menu independent N and its induced μ . Intuitively, the set $\overline{H}(a) \cap A$ contains exactly those items in A that are more salient than a . The Axiom 6 requires that the probability of choosing a more salient item than $a \in \text{supp}(A)$ only depends on the size of $\overline{H}(a) \cap A$.

Axiom 6. (Independence of More Salient Choices)

Let $t \in \mathcal{T}$ and two menus A, B be arbitrary. For any $a \in \text{supp}(A), b \in \text{supp}(B)$, $|\overline{H}(a) \cap A| \geq |\overline{H}(b) \cap B| \iff \sum_{c \in \overline{H}(a) \cap A} \rho(c, A, t) \geq \sum_{d \in \overline{H}(b) \cap B} \rho(d, B, t)$.

Proposition 4. (*Gradual Attention Representation*)

Under the assumption of time variation, ρ has a gradual attention representation with menu independent $(\succ, R, N)$ if and only if ρ satisfies Axioms 1-6. Moreover, $\succ, R$ is unique and the μ induced by the counting process N is unique up to $\max_{A \in \mathcal{X}, a \in \text{supp}(A)} |\overline{H}(a) \cap A|$.

3 Comparative Statics

In this section, we give some comparative statics results. Note that the Proposition 2 itself is a comparative statics result and the result is modified as the Axiom 1 and 5 in our main characterization. Since we can interpret the counting process as a measure of the DM's learning/searching speed, it is natural to compare the choice behaviors for two DMs with different speeds.

Proposition 5. (*Comparative Speed*)

Let two DMs share the same $\succ, R$ but different counting processes N^1, N^2 such that the speed of the first DM is stochastically slower, i.e., $\mu_t^{N^1} \succ_F \mu_t^{N^2}$ w.r.t the natural order for all t . Then the second DM is making better choice at any time t for any menu A , i.e., $\rho_1(\cdot, A, t) \succ_F \rho_2(\cdot, A, t)$ w.r.t $\succ$.

The proof follows immediately from the Proposition 2 and is omitted. In particular, if these two DMs follow the Poisson process with $\lambda_2 > \lambda_1$, then it will satisfy the definition of the comparative speeds above. Next we consider two DMs with the same $\succ, N$ but with different salience orders. The only reason that a DM fails to choose the $\succ$ -best item in our model is because it is not salient enough such that the DM can pay attention to it by the deadline. Thus we should expect that if the preference and the salience orders are more aligned, the DM will make better choices. We formally define the alignment with the preference as follows.

Definition 3. (Comparison of Alignment with Preference)

Given $R_1, R_2, \succ \in \mathcal{S}$, R_1 is said to be *more aligned with $\succ$* than R_2 , if for any $a \succ b$, $aR_2b \implies aR_1b$. We denote $R_1 >_\succ R_2$ and it is clear that $>_\succ$ is a partial order, i.e., reflexive, anti-symmetric and transitive.

Corollary 2. (*Comparative Alignment with Preference and Accuracy*)

Suppose two DMs share the same $\succ, N$ but with different R_1, R_2 and R_1 is more aligned with $\succ$ than R_2 , then for any menu A and time t , $\rho^1(\cdot, A, t) \succ_F \rho^2(\cdot, A, t)$ w.r.t $\succ$.

If the salience order is more aligned with the preference, then more items will be dominated and thus the support of ρ will be smaller. It turns out the converse is also true. We provide an equivalent characterization of comparative alignments of the salience orders w.r.t the preference and the set inclusion relation over the support of ρ as the following proposition.

Proposition 6. (*Comparative Alignment with Preference and Support Inclusion*)

Let ρ_1, ρ_2 have gradual attention representations with $(\succ_1, R_1, N_1), (\succ_2, R_2, N_2)$ such that $\succ_1 = \succ_2 = \succ$. Then the following two statements are equivalent:

1. R_1 is more aligned with $\succ$ than R_2 , i.e., $R_1 \succ_{\succ} R_2$.

2. For any menu $A \subseteq X$, $\text{supp}_1(A) \subseteq \text{supp}_2(A)$.

Similarly, we can compare two preferences in terms of their alignments with a given salience order. We obtain a dual characterization of comparative alignments of the preferences w.r.t the salience order and the set inclusion relation over the support of ρ .

Definition 4. (Comparison of Alignment with Salience)

Given $\succ_1, \succ_2, R \in \mathcal{S}$, $\succ_1$ is said to be *more aligned with R* than $\succ_2$, if for any aRb , $a \succ_2 b \implies a \succ_1 b$. We denote $\succ_1 \succ_R \succ_2$ and it is clear that $\succ_R$ is a partial order, i.e., reflexive, antisymmetric and transitive.

Proposition 7. (Comparative Alignment with Salience and Support Inclusion)

Let ρ_1, ρ_2 have gradual attention representations with $(\succ_1, R_1, N_1), (\succ_2, R_2, N_2)$ such that $R_1 = R_2 = R$. Then the following two statements are equivalent:

1. $\succ_1$ is more aligned with R than $\succ_2$, i.e., $\succ_1 \succ_R \succ_2$.

2. For any menu $A \subseteq X$, $\text{supp}_1(A) \subseteq \text{supp}_2(A)$.

The Propositions 6 and 7 will be quite useful when we discuss the aggregation property of our model, i.e., allowing heterogeneity in primitives.

4 Lack of Variations

The rich data requirement the gradual attention model might be infeasible in the applications. In this section, we discuss the gradual attention model without choice variations either across menus or over time. In the former case, we provide necessary and sufficient conditions to characterize a menu dependent gradual attention representation. In absence of choice variations across menus, we show identification of the primitives is typically partial, but still quite straightforward. In the latter case, we also show identification of the primitives in absence of variations over time is generically partial, though a mild *regularity* assumption will lead to full identification. Note that the domain of the choice data without time variations is used in the other existing stochastic choice models in the literature. We discuss some connections between these models and the gradual attention model without time variations.

4.1 Without Variation across Menus

By lack of variations across menus, we mean only choices for some menu A over all time $t \in \mathcal{T}$ are observed, i.e., $\rho(\cdot, A, t)$ for all $t \in \mathcal{T}$. As has been shown in the Proposition 2, a key feature of the gradual attention model is that the DM will make better choices over time in terms of the FOSD w.r.t the $\succ$. It turns out a modified version of this feature is also sufficient for characterizing the menu-dependent gradual attention model. The reason we need to modify it is because identification of the $\succ$ may not be possible and there could be multiple candidate preference orders $P \in \mathcal{S}$ such that better choices are made over time in terms of the FOSD w.r.t P .

Proposition 8. (*Menu-Dependent Gradual Attention Representation*)

Let a menu A be arbitrary, then $\{\rho(\cdot, A, t)\}_{t \in \mathcal{T}}$ has a menu-dependent gradual attention representation, i.e., $(\succ_A, R_A, N^A)$ depends on the menu A , if and only if there exists $P \in \mathcal{S}$ such that $\rho(\cdot, A, t) \succ_F \rho(\cdot, A, s)$ w.r.t $P \in \mathcal{S}$ for all $t > s$. Moreover, the following identification statements about $P, \succ_A, R$ are true:

- *P is aligned perfectly with $\succ_A$ and is misaligned perfectly with R over $\text{supp}(A)$*
- *Suppose $a \in \text{supp}(A), b \in A \setminus \text{supp}(A)$, then $aPb, a \succ_A b, aRb$.*

Note that the condition “there exists $P \in \mathcal{S}$ such that $\rho(\cdot, A, t) \succ_F \rho(\cdot, A, s)$ w.r.t $P \in \mathcal{S}$ for all $t > s$ ” is equivalent to the expected utility of any DM who has the same preference as P is increasing over time, regardless of the risk attitudes. We believe this interpretation of the characterization result has more normative appealing. In the section of experimental evidence, the nature of the experiment disables us from exploiting choice variations across

menus and the only testable implication is then the increasing accuracy in terms of the FOSD.

4.2 Without Variations over Time: A Static Model

By lack of variations over time, we mean choices across all menus but only for some time t are observed, i.e., $\rho(\cdot, A, t)$ for all $A \in \mathcal{X}$. The reason we focus on the same t for all menus is for simplicity. We will show even for this simple case, identifying the primitives of the model can be quite difficult. Thus we consider the random choice rule without the time dimension but we still use the same notation $\rho : X \times \mathcal{X} \rightarrow [0, 1]$ for convenience.

In most models of stochastic choice with limited attention, the primitives contain a preference and a probability measure over the set of possible consideration sets for each menu. And choices are then made from the realized consideration sets based on the preference. Unlike these models, we model more details on the formation of the consideration set. The primitives of our static model are the same as the general model, except that we only have one probability distribution μ over positive integers that determines the size of the consideration set. We focus on μ s such that $\mu(Z) > 0$ for all $Z \leq |X|$. This additional restriction on μ will make the model more tractable. Formally, the primitives are $(\succ, R, \mu)$ and we say ρ has a gradual attention representation without time variation if for all a, A :

$$\rho(a, A) = \sum_{\{Z \in \mathbb{Z}^{++} : \mathcal{C}_{\succ}(A^R(Z)) = a\}} \mu(Z)$$

We discuss some identification issues, i.e., recovering the primitives $(\succ, R, \mu)$ by observing the data ρ . We claim that full identification of $(\succ, R, \mu)$ is generally not possible. The easiest way to see this is to consider a case when there are only two items a, b and we observe $\rho(a, \{a, b\}) \in (0, 1)$. Then it can be easily verified that the following two different sets of primitives both generate ρ ,

$$\begin{aligned} a \succ b \quad bRa \quad \mu(Z=1) &= \rho(a, \{a, b\}) \\ b \succ' a \quad aR'b \quad \mu'(Z=1) &= 1 - \rho(a, \{a, b\}) \end{aligned}$$

Note in contrast, by Proposition 1, assessing the choice dynamics over time will easily bring us identifications of $\succ, R$. In order to resolve this identification problem, we start by considering a necessary and sufficient condition for the identification of $\succ, R$ when there are only three items. This simple situation will shed lights on recovering $\succ, R$ and understanding the choice reversals across menus in more general settings.

Lemma 2. (*Identification of $\succ, R$ when $|X| = 3$*)

Suppose $|X| = 3$ and ρ has a gradual attention representation without time variation, then $\succ, R$ can be identified uniquely if and only if either of the following two conditions hold:

1. $\rho(\cdot, A) \in \{0, 1\}$ for all $A \subset X$ such that $|A| = 2$.
2. Positivity (i.e., $\rho(a, A) > 0$ for all a, A) and $\rho(\cdot, \{a, b\}) \neq \frac{1}{2}$ for all $a, b \in X$.

We discuss how $\succ, R$ can be identified when the condition 2 is met. Note that $\text{supp}(X) = X$ is equivalent to saying $\succ, R$ are perfectly misaligned. Consider what happens if adding item c to the doubleton menu $\{a, b\}$. If c is more salient than both a, b , it must be the case that the choice probabilities of both a, b decrease. If c is less salient than the more salient of a, b , say a , then the choice probability of a will not be affected while the choice probability of b will decrease. These two possibilities are exhaustive and can be summarized as follows:

- $\rho(a, ab) = \rho(a, abc) \iff aRb, aRc, b \succ a, c \succ a$
- $\rho(a, ab) < \rho(a, abc) \iff cRa, a \succ c$

Thus we can first find the most salient (the least preferred) item, say a , whose choice probability in any doubleton menu will not be affected whenever a third item is added. Then we consider a doubleton menu containing a , say $\{a, b\}$, it must be the case that $\rho(a, ab) = \mu(1)$. We then check $\rho(b, bc)$ and it is either equal to $\mu(1)$ or $\mu(\geq 2)$ but not both. This is because the condition 2 rules out the possibility that $\rho(b, bc) = \mu(1) = \mu(\geq 2) = \frac{1}{2}$. If $\rho(b, bc) = \rho(a, ab)$, then we have $bRc, c \succ a$. If $\rho(b, bc) = 1 - \rho(a, ab)$, then we have the converse. This completes the identification.

Here a natural question to ask is when the necessary and sufficient conditions fail, to what extent we can identify $\succ, R$? To address this question, we give a formal definition by which we mean two items are not identified by $\succ, R$. When two items are not identified by $\succ, R$, we say they are “equivalent” (in terms of identification).

Definition 5. (Equivalent Items)

Given ρ that has a gradual attention representation without time variation, two items a, b are said to be *equivalent*, (or not identified), if there exists $\succ_1, R_1$ and $\succ_2, R_2$ such that

1. $\succ_1, R_1, \mu_1$ and $\succ_2, R_2, \mu_2$ both generate ρ for some μ_1, μ_2 .
2. $a \succ_1 b, bR_1a$ and $b \succ_2 a, aR_2b$.

When a, b are equivalent, we denote $a \sim b$.

It is not immediately clear that $\sim$ is an equivalence relation. But we can (vacuously) define $c \sim c$ for all $c \in X$ and we will show for any $a \neq b \neq c$, it cannot be $a \sim b$ and $b \sim c$. It follows that transitivity holds vacuously and therefore $\sim$ is an equivalence relation.

Lemma 3. *(Equivalence Class)*

$\sim$ is an equivalence relation and for any item $a \in X$, its equivalence class $[a]$ contains at most two elements.

The above the Lemma 1 shows that we can uniquely partition the X into a collection of equivalent classes according to $\sim$ and each equivalent class contains at most two items. By the restriction of $\succ$ (and similarly for R) over two equivalent classes $[a], [b]$, we define $[a] \succ [b]$ if and only if $c \succ d$ for any $c \in [a], d \in [b]$. We shall show that the restriction of $\succ, R$ on the equivalent classes must be unique. We summarize the general partial identification result as the following theorem.

Theorem 2. *(Partial Identification)*

Suppose ρ has a gradual attention representation without time variation, then $\succ, R$ are identified uniquely up to its equivalence classes. That is, there exists a unique collection of equivalence classes by $\sim$ such that the collection of equivalence classes forms a partition of X and the restriction of $\succ, R$ on the collection of equivalence classes is unique. Moreover, each equivalence class contains at most two items.

The above partial identification results highlight the importance of utilizing the choice variations over time to obtain full identification. We also propose a sufficient condition such that full identification is generically feasible.

Proposition 9. *(Generic Conditional Identification)*

Suppose ρ has a gradual attention representation without time variation and $\max_A |\text{supp}(A)| \geq 3$. Then $\succ, R$ can be generically identified uniquely, i.e., the set of ρ s that fails full identification has Lebesgue measure zero.

The sufficient condition $\max_A |\text{supp}(A)| \geq 3$ itself is quite likely to be satisfied when $|X|$ is large as the following result.

Lemma 4. *(Asymptotic Support Requirement)*

As $|X|$ goes large, the probability that the pair of $\succ, R$ satisfying the $\max_A |\text{supp}(A)| \geq 3$ converges to 1.

To summarize, identifications of the primitives without time variations may not always be possible. But we can still achieve partial identifications, partitioning the items into cells containing two items and achieving unique identification of $\succ, R$ restricted over the partition. On the other hand, when the collection of items is large enough, the chance of full identification is converging to one.

5 Heterogeneity and Representative Agent

In this section, we discuss heterogeneity of the primitives $(\succ, R, N)$ in the gradual attention model. Since the stochastic choice data often comes from the aggregated choice data of the population, it will add more flexibility to allow heterogeneity within the population. We consider the aggregation property of the gradual attention model by allowing heterogeneity in each of the three primitives $(\succ, R, N)$, while assuming homogeneity in the other two.

Recall that we characterize the gradual attention model with menu-dependent counting processes in the Theorem 1, while in the Proposition 4 we proposed the Axiom 6 to restricting the counting process to be menu independent. In this section, we will focus on aggregation properties of the gradual attention model with menu dependent counting process. We will first show that the gradual attention model is closed under aggregation over the counting processes, i.e., there exists a representative agent behaving according to the model. Note that this result is also true when the counting process is menu independent. We next show that the gradual attention model is closed under aggregation over the salience orders if and only if the set of salience orders can be totally ordered according to their alignments with the preference. Finally, we use examples to demonstrate that a representative agent does not exist when allowing heterogeneity in the preferences.

We model heterogeneity by a probability measure over the set of each of the primitives, $m^P \in \Delta\mathcal{S}$ for the preferences, $m^S \in \Delta\mathcal{S}$ for the salience orders, and $m^C \in \Delta\mathcal{N}$. For each of the probability measure, we denote its support by $\text{supp}(m^P) =: \{\succ \in \mathcal{S} : m^P(\succ) > 0\}$ (similarly for $\text{supp}(m^S), \text{supp}(m^C)$).⁸ If a representative agent exists, then we will use $\bar{\succ}, \bar{R}, \bar{N}$ (and $\bar{\mu}$, which is induced by $\bar{N}$) to denote the representative primitives.

5.1 Heterogeneity of the Counting Process

We start by considering heterogeneity in the counting process so ρ is now determined by $\succ, R, m^C$. For each $N \in \text{supp}(m^C)$, we denote the random choice rule generated by $\succ, R, N$ as ρ_N . We say a representative agent exists if $\rho = \int \rho_N dm^C(N)$ can be generated by some $\bar{\succ}, \bar{R}, \bar{N}$.

It is clear that we need to aggregate μ_t^N , the induced probability measures over $\mathbb{Z}^{++}$ by each $N \in \text{supp}(m^C)$, as follows: for each $t \in \mathcal{T}, Z \in \mathbb{Z}^{++}$, the probability that a DM's

⁸Not to be confused with the support of a menu for a random choice rule, though we use the same notation $\text{supp}()$.

consideration set has size equal to Z becomes:

$$\bar{\mu}_t(Z) = \int \mu_t^N(Z) dm^C$$

Lemma 5. (*Aggregation of Counting Processes*)

There exists a counting process $\bar{N} \in \mathcal{N}$ that induces $\bar{\mu}_t$.

Proposition 10. (*Representative Agent under Heterogeneous Counting Processes*)

Suppose ρ is generated by heterogeneous agents with $\succ, R, m^C$, then there exists a representative agent $\bar{\succ} = \succ, \bar{R} = R, \bar{N}$.

The above proposition follows immediately from the lemma and the proof is omitted. The main takeaway of this proposition is that the heterogeneity in the counting process will not affect the identification of the $\succ, R$, as well as all the testable implications, i.e., the Axioms 1-5. In particular, this proposition allows us to test the increasing accuracy implication in the section of experimental evidence, when the participants of the experiment are expected to have different evaluation speeds.

5.2 Heterogeneity of the Saliency Order

We next consider heterogeneity in the saliency orders so ρ is now determined by $\succ, m^S, N$. For each $R \in \text{supp}(m^S)$, we denote the random choice rule generated by $\succ, R, N$ as ρ_R . We say a representative agent exists if $\rho = \int \rho_R dm^S(R)$ can be generated by some $\overline{\succ}, \overline{R}, \overline{N}$, where we again allow $\overline{N}$ to be menu dependent while $\overline{\succ}, \overline{R}$ are restricted to be menu independent.

Recall that we have defined the “more alignment” $>_{\succ}$ w.r.t a given $\succ$ and the set $\mathcal{S}$ is partially ordered by $>_{\succ}$. We first use a simple example to show that if there exists two elements of $\text{supp}(m^S)$ such that they cannot be compared by $>_{\succ}$, then a representative agent cannot exist.

Example 1. (Nonexistence of Representative Agent under Heterogeneous Saliency)

Suppose $a \succ c \succ b$ and there are two types of saliency orders aR_1bR_1c, cR_2aR_2b .

It is easy to check that R_1 is more aligned with $\succ$ over a, c but R_2 is more aligned with $\succ$ over b, c and therefore they cannot be ordered by $>_{\succ}$. Now suppose a representative agent exists, then it can be easily verified that $\overline{\succ} = \succ$. Consider each doubleton menu and note that $\text{supp}(\{a, c\}) = \{a, c\}, \text{supp}(\{b, c\}) = \{b, c\}, \text{supp}(\{a, b\}) = a$ since cR_2a, bR_1c . It follows $c\overline{R}a\overline{R}b\overline{c}$, a contradiction to $\overline{R} \in \mathcal{S}$.

It turns out a necessary and sufficient condition for a representative agent with $\overline{\succ}, \overline{R}$ to exist is that the elements of $\text{supp}(m^S)$ can be totally ordered by $>_{\succ}$. When such a representative agent exists, $\overline{R} \in \text{supp}(m^S)$ is less aligned with $\succ$ than any elements of $\text{supp}(m^S)$, i.e., the minimal element of $\text{supp}(m^S)$.

Proposition 11. (Gradual Attention with Saliency Heterogeneity)

Suppose ρ is generated by the gradual attention model with saliency heterogeneity $(\succ, m^S, N)$, then there exists a representative agent with $(\overline{\succ}, \overline{R}, \overline{N})$ if and only if the elements of $\text{supp}(m^S)$ can be totally ordered by $>_{\succ}$. Moreover, $\overline{R}$ is the minimal element of $\text{supp}(m^S)$, i.e., $\overline{R} \in \text{supp}(m^S)$ and $R >_{\succ} \overline{R}$ for any $R \in \text{supp}(m^S)$.

Note that in the Example 1, we generate contradictions in terms of the support compatibility conditions across menus, i.e., the Axiom 4 is violated in particular. In a situation when we lack of variations across menus, the Axiom 4 will no longer be a concern since the only testable implication becomes the increasing accuracy as the Proposition 8 shows. The following corollary shows a menu-dependent gradual attention model is closed under arbitrary heterogeneity of saliency orders.

Corollary 3. (Menu-dependent Gradual Attention with Saliency Heterogeneity)

Fix an arbitrary menu A . Suppose $\{\rho(\cdot, A, t)\}_{t \in \mathcal{T}}$ is generated by a menu-dependent gradual

attention model with salience heterogeneity $\succ, m^S, N$, then there exists a representative agent with $\overline{\succ} = \succ, \overline{R}, \overline{N}$.

This corollary will allow us to test the increasing accuracy implication within a menu in the section of experimental evidence, when the participants of the experiment are expected to evaluate the items according to arbitrarily different salience orders.

5.3 Heterogeneity of the Preference

Finally we consider heterogeneity in the preferences so ρ is now determined by m^P, R, N and again we allow the counting process N to be menu dependent. For each $\succ \in \text{supp}(m^P)$, we denote the random choice rule generated by $\succ, R, N$ as $\rho_\succ$. We say a representative agent exists if $\rho = \int \rho_\succ dm^P(\succ)$ can be generated by some $\bar{\succ}, \bar{R}, \bar{N}$, where we again allow $\bar{N}$ to be menu dependent while $\bar{\succ}, \bar{R}$ are restricted to be menu independent.

We shall first prove a dual result of the Corollary 3, which says the increasing accuracy implication for a given menu is closed under arbitrary heterogeneity of preferences, while we need to modify the “accuracy” since there are multiple preferences.

Corollary 4. (*Menu-dependent Gradual Attention with Preference Heterogeneity*)

Fix an arbitrary menu A . Suppose $\{\rho(\cdot, A, t)\}_{t \in \mathcal{T}}$ is generated by a menu-dependent gradual attention model with preference heterogeneity m^P, R, N , then there exists a representative agent with $\bar{\succ}, \bar{R} = R, \bar{N}$.

Proposition 12. (*Gradual Attention with Preference Heterogeneity*)

Suppose ρ is generated by the gradual attention model with salience heterogeneity (m^P, R, N) , then there exists a representative agent with $(\bar{\succ}, R, \bar{N})$, where $\bar{\succ}, R$ are menu independent, if and only if the elements of $\text{supp}(m^P)$ can be totally ordered by $>_R$. Moreover, $\bar{\succ}$ is the minimal element of $\text{supp}(m^P)$, i.e., $\bar{\succ} \in \text{supp}(m^P)$ and $\bar{\succ} >_R \bar{R}$ for any $R \in \text{supp}(m^P)$.

The Proposition 12 is a dual result of the Proposition 11 and their proofs are similar.

To summarize,

- A representative agent always exists under arbitrary heterogeneity of counting processes.
- Without menu variations, a representative agent exists under arbitrary heterogeneity of the primitives.

6 Experimental Evidence

In this section, we revisit the experimental data in Caplin, Dean, and Martin (2011) and use it to test the increasing accuracy implication of our gradual attention model. We do not test the rest of the implications because testing these implications requires observing choice variations across menus and the data do not contain such variations. We found suggestive evidence in support of the increasing accuracy choice pattern.

6.1 Data Description

We briefly describe the experiment by Caplin, Dean, and Martin (2011). An experimental session contains multiple rounds and in each round, a participant was asked to make a choice by a certain amount of time (120 seconds). A choice set contains several elementary algebraic expressions, whose evaluation determines the monetary payoffs. For example, a typical expression can be “nine minus two plus four minus three” and it is evaluated to be equal to 9. The math operators are either plus or minus. Both the math operators and the integers are intentionally presented in English to make evaluation tasks more difficult. During a session, a participant will choose from a variety choice sets with different sizes and complexities (in terms of number of math operators). The following Figure 1 that is borrowed from Caplin, Dean, and Martin (2011)) gives an example of a typical round.

Round 2 of 30

Current selection: four plus eight minus four

Choose one:

☐	zero	0
☐	three plus five minus seven	1
☐	four plus two plus zero	6
☐	four plus three minus six	1
☐	four plus eight minus four	8
☐	three minus three plus one	1
☐	five plus one minus one	5
☐	eight plus two minus five	5
☐	three plus six minus five	4
☐	four minus two minus one	1
☐	five plus five minus one	9

Finished

Figure 1: A Typical Choice Round (Figure 1 in Caplin, Dean, and Martin (2011))

At the start of a choice round, a participant is endowed with a default option with zero

monetary payoff and can update their choice at any time before the 120 seconds expires. At the end of a experimental session, a random round and a random time point will be selected by the computer for this participant. The participant receives a monetary prize that is proportional to the value of her choice at the selected time in the selected round. This rewarding mechanism provides the participant incentives to update her choice instantly whenever a more valuable option is discovered.

The data collected by Caplin, Dean, and Martin (2011) is called the “choice process data”. Each observation of the data includes the entire choice switching history for one participant in one round. One can easily show, within a round, a participant is making better choices as time increases if and only if this participant never switches to a worse choice. Instead of focusing on individual level deterministic choice switches, we consider aggregated choice data for each menu by different participants at different time points. Formally, for each observation, the data we use is the randomly selected time point and the participant’s choice at that time point ⁹. We end up with 13902 observations of individual level choice data, after we drop 670 observations for practice rounds.

We next aggregate the individual level choice data into choice data for each menu. It is challenging to identify a menu because the reported data display the evaluations of options as a vector of non-negative integers, instead of the original algebraic expressions. We treat two menus as the same one if the two vectors are the same. This treatment might mistakenly classify two different menus as the same one, though we believe it is unlikely that two different menus will have the same vector of integers as valuations of their options. A menu might contain multiple options with the same evaluations and we consider these duplicated options as the same option. We end up with 178 menus and the number of choice observations for each menu ranges from 29-156, with a median equal to 72.

For each menu, we are going to test whether choices are better as time increases. Due to the lack of observations, we do not use the original precision of time in seconds. For each menu, we construct two time bins (high/low) such that around half of the observations fall into each time bin. The low time bin contains choices that are made with less time than those in the high time bin.

⁹In their released data, Caplin, Dean, and Martin (2011) did not report the entire choice history as a time series. Instead, only the computer randomly selected time point and the associated choice were reported in each observation.

6.2 Tests and Results

Within a menu, choices in each time bin can be normalized as a probability distribution, i.e., stochastic choice. For menu i , denote F_i^H (F_i^L) as the empirical cumulative distribution function (CDF) for the high (low) time bin. We say menu i exhibits the empirical FOSD choice pattern if $F_i^H(x) \leq F_i^L(x)$ for all x in their support. We test the empirical FOSD choice pattern for all menus and 122 of the 178 menus exhibit this choice pattern. The following Figure 2 shows two menus with/without the empirical FOSD choice pattern.

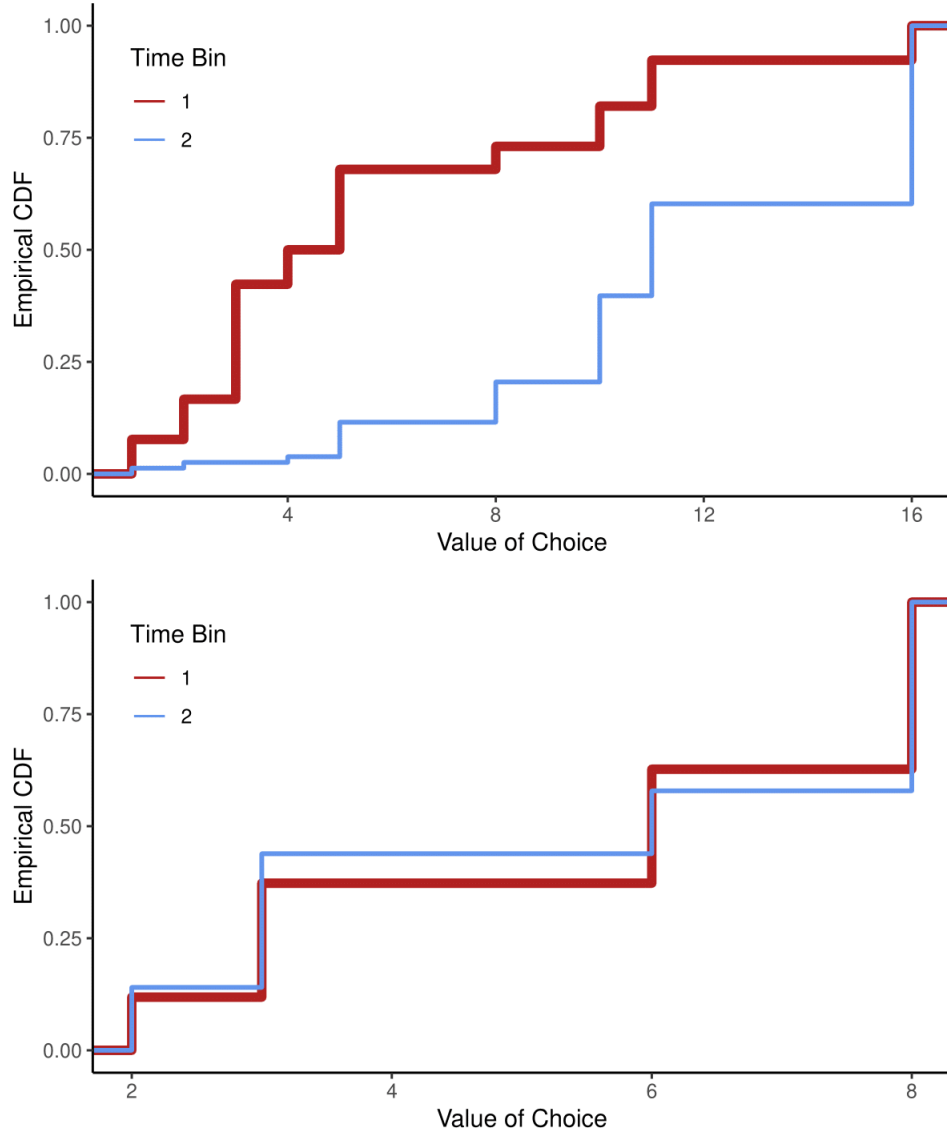


Figure 2: Menus with/without the empirical FOSD choice pattern

We next conduct statistical tests for first-order stochastic dominance. For each menu i , we test the null hypothesis H_0 that the choices from the high time bin first-order stochastically

dominates choices from the low time bin against its alternative hypothesis H_1 .

$$H_0 : F_i^H(x) \leq F_i^L(x) \text{ for all } x$$

$$H_1 : F_i^H(x) > F_i^L(x) \text{ for some } x$$

We adopt the test proposed by Barrett and Donald (2003). The Barrett-Donald test is a supremum-type test and the test statistic is given by (N_i^H, N_i^L are the sample sizes for high and low time bins in menu i)

$$\sqrt{\frac{N_i^H N_i^L}{N_i^H + N_i^L}} \cdot \max_x (F_i^H(x) - F_i^L(x))$$

The confidence interval of the test statistics is generated by bootstrap under the least favorable case of the null hypothesis H_0 (i.e., $F_i^H = F_i^L$). The p-values of the test statistics for all menus are summarized in Figure 3.

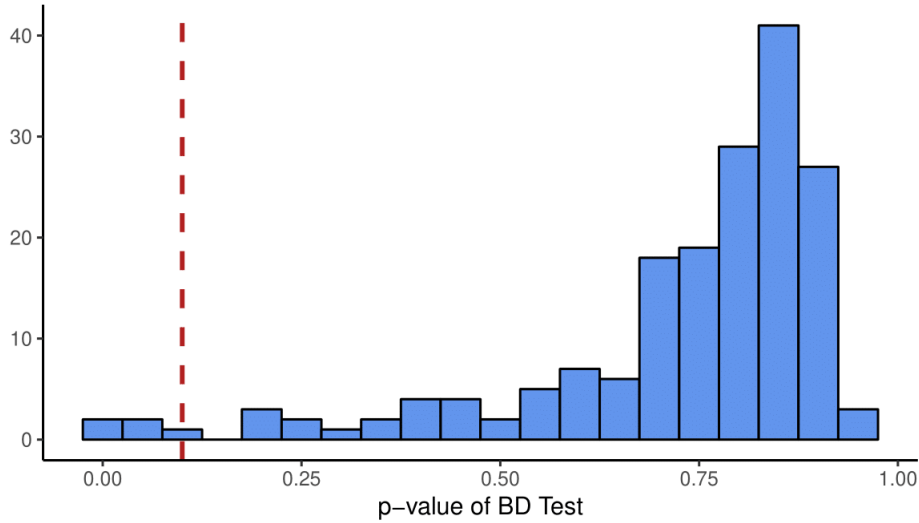


Figure 3: The distribution of p-values of Barrett-Donald test

Figure 3 gives the distribution of p-values of the Barrett-Donald test statistics for all menus. The red vertical line indicates the 10% critical level. Even at the level of 10%, we can reject the null for only 5 out of the 178 menus. There is only 1 menu for which we can reject the null at 1% level.

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7 Appendix

7.1 Proofs in the main text

Proof. (Proof of Proposition 1) The claim that $\Delta_{ab}(t)$ is constant if and only if $\Delta_{ab}(t) = 1$ or 0 follows immediately from the time variation assumption. We first show $\Delta_{ab}(t)$ is a monotone function in t . There are four possibilities of $\succ, R$ on a, b . If $a \succ b, aRb$ ($b \succ a, bRa$), $\Delta_{ab}(t)$ is a constant function of t , equal to 1 (-1) since the item that is less salient and less preferred will never be chosen.

If $a \succ b, bRa$, $\Delta_{ab}(t)$ cannot be constant. This is because the second property of the counting process implies there exists some t_1, t_2 such that $\mu_{t_1}^N(1) > 0, \mu_{t_2}^N(\geq 2) > 0$, which implies $\Delta_{ab}(t_1) < 1, \Delta_{ab}(t_2) > -1$. By the time variation assumption, $\Delta_{ab}(t)$ then cannot be constant. $\Delta_{ab}(t)$ will be increasing in t since the probability that the consideration set being singleton is decreasing in t and hence the choice probability of the more preferred item will increase. If $b \succ a, aRb$, a symmetric argument shows $\Delta_{ab}(t)$ must be decreasing in t . We conclude that $\Delta_{ab}(t)$ must be monotonic in t in each of these four possible cases. The $\iff$ statements for the four observations follow immediately since these four possibilities are exhaustive.

The above argument also concludes the only if part for the characterization of the representation when $|X| = 2$. It remains to show the if part. If $\Delta_{ab}(t)$ is constant, then any counting process work. If $\Delta_{ab}(t) \uparrow$ (similarly for $\Delta_{ab}(t) \downarrow$), let $\mu(1) = 1 - \Delta_{ab}(t) \uparrow, \mu(2) = \Delta_{ab}(t) \uparrow$. By the Lemma 7, we can find a desirable counting process that induces μ . $\square$

Proof. (Proof of Proposition 2) $t \geq s \implies \mu_t^N \succ_F \mu_s^N$ (w.r.t the natural order $>$ over $\mathbb{Z}^+$) by the Lemma 6. Also $A^R(Z') \subseteq A^R(Z)$ for all $Z' < Z$ by the construction of the consideration set. Finally, for all $B' \subseteq B$, either $\mathcal{C}_\succ(B) \succ \mathcal{C}_\succ(B')$ or $\mathcal{C}_\succ(B) = \mathcal{C}_\succ(B')$. Combining these facts yield the result since for any $a \in A$,

$$\begin{aligned} \sum_{\{b \in A: a \succ b\}} \rho(b, A, s) &= \sum_{\{b \in A: a \succ b\}} \sum_{\{Z \in \mathbb{Z}^{++}: \mathcal{C}_\succ(A^R(Z))=b\}} \mu_s^N(Z) \geq \\ &\sum_{\{b \in A: a \succ b\}} \sum_{\{Z \in \mathbb{Z}^{++}: \mathcal{C}_\succ(A^R(Z))=b\}} \mu_t^N(Z) = \sum_{\{b \in A: a \succ b\}} \rho(b, A, t) \end{aligned}$$

$\square$

Proof. (Proof of Corollary 1) By the representation 1, $\rho(\mathcal{C}_\succ(A), A, t) \geq \mu_t(\geq |A|)$ for any menu A . For any $\epsilon > 0$, let the K in the assumption be $|A|$, then there exists s such that $\mu_s(< |A|) < \epsilon \iff \mu_s(\geq |A|) \geq 1 - \epsilon$. Due to the non-decreasing property of the counting process, this means for all $t \geq s$, $\rho(\mathcal{C}_\succ(A), A, t) \geq \mu_t(\geq |A|) \geq \mu_s(\geq |A|) \geq 1 - \epsilon$. $\square$

Proof. (Proof of Lemma 1) Let a impact b and b dominate c . If c dominates a , then the Axiom 2 implies b dominates a , a contradiction. If c impacts a , then the Axiom 3 implies c impacts b , a contradiction. The Axiom 1 implies these two cases along with the two cases in the first part of the statement are exhaustive. A similar argument can be used to prove the second part of the statement. $\square$

Proof. (Proof of Theorem 1) The only if part: Axiom 1-3 follow from the Proposition 1. The Axiom 4 comes immediately from the representation. The Axiom 5 follows from the Proposition 2 since the $\overline{M}(a)$ coincides with the upper contour set of a w.r.t the $\succ$.

We prove the if part by construction.

Step 1. (Construct menu-independent $\succ, R \in \mathcal{S}$ from ρ over doubleton menus.)

Construct $\succ$ by $a \succ b \iff a$ dominates b or a impacts b (i.e., $\Delta_{ab}(t) \uparrow$ or $\Delta_{ab}(t) = 1$ for all $t \in \mathcal{T}$). Construct R by $aRb \iff a$ dominates b or b impacts a (i.e., $\Delta_{ab}(t) \downarrow$ or $\Delta_{ab}(t) = 1$ for all $t \in \mathcal{T}$). The Axiom 1 guarantees we can do such constructions over any pair of a, b by their doubleton menu. We now verify $\succ, R \in \mathcal{S}$. It remains to check transitivity since the completeness and the anti-symmetry follow from the construction.

Let $a \succ b, b \succ c$. If a dominates b , b dominates c (or a impacts b , b impacts c), then the Axiom 2 (or the Axiom 3) implies a dominates c (or a impacts c), it follows $a \succ c$. If a dominates b , b impacts c (or a impacts b , b dominates c), then by the Lemma 1, a dominates or impacts c and thus $a \succ c$.

Let aRb, bRc . If a dominates b , b dominates c (or a is impacted by b , b is impacted by c), then the Axiom 2 (or the Axiom 3) implies a dominates c (or a is impacted by c), it follows aRc . If a dominates b , b is impacted by c . Suppose cRa so either c dominates a or a impacts c . But c dominates a and a dominates b implies c dominates b by the Axiom 2, which is a contradiction to c impacts b . On the other hand, a impacts c and c impacts b implies a impacts b by the Axiom 3, which is a contradiction to a dominates b . Thus we conclude aRc . The other case when a impacts b , b dominates c can be proved symmetrically.

Step 2. (Verify the support of ρ is compatible with the constructed $\succ, R$.) For an arbitrary menu A , we verify the $\text{supp}(A)$ is compatible with the constructed $\succ, R$, i.e., $a \in A \setminus \text{supp}(A) \iff$ there exists some $b \in A$ such that $b \succ a, bRa$. But by the construction, $b \succ a, bRa$ implies b dominates a . The compatibility requirement is then satisfied by the Axiom 4.

Step 3. (Construct a (menu dependent) counting process N .) For an arbitrary menu A , we can label the alternatives in $\text{supp}(A)$ (say $|\text{supp}(A)| = J$) as $\text{supp}(A) = \{a^1, a^2, \dots, a^J\}$ such that $a^j \succ a^i \iff j > i$. Since the constructed $\succ, R$ satisfies the support compatibility condition, the restriction of $\succ, R$ over $\text{supp}(A)$ must be perfectly misaligned. It follows $a^iRa^j \iff j > i$.

For each a^j , define a function by $\phi(a^j) := |\{a \in A : aRa^j\}|$. Note that by construction, $\phi(a^1) = 0$ and for $i < j$, $\phi(a^j) - \phi(a^i)$ is the size of the set of items that are less salient than a^i and more salient than a^j , but are never chosen. Now for each $t \in \mathcal{T}$, we construct $\mu_t(Z)$ as follows:

- If $J = 1$. For any $Z \in \mathbb{Z}^{++}$, $1 \leq Z \leq |X|$, let $\mu_t(Z) = 1/|X|$
- If $J \geq 2$. For each $j \in \{1, 2, \dots, J-1\}$, let $\mu_t(Z) = \rho(a^j, A, t)/(\phi(a^{j+1}) - \phi(a^j))$ for all $Z \in \mathbb{Z}^{++}$ such that $\phi(a^j) \leq Z \leq \phi(a^{j+1})$. For $\phi(a^J) < Z \leq |X|$, let $\mu_t(Z) = \rho(a^J, A, t)/(|X| - \phi(a^J))$.

We prove the following **claim**: $\mu_t \succ_F \mu_s$ w.r.t the natural orders on $\mathbb{Z}^{++}$ for any $t \geq s$.

Proof. We want to prove for any $1 \leq k \leq |X|$, $\sum_{1 \leq Z \leq k} \mu_t(Z) \leq \sum_{1 \leq Z \leq k} \mu_s(Z)$. When $J = 1$, this is trivially true by construction since $\mu_s(Z) = \mu_t(Z) = \frac{1}{|X|}$. Suppose $J \geq 2$. If $\phi(a^J) < k \leq |X|$,

$$\begin{aligned} \sum_{1 \leq Z \leq k} \mu_t(Z) &\leq \sum_{1 \leq Z \leq k} \mu_s(Z) \iff \\ \sum_{1 \leq Z \leq \phi(a^J)-1} \mu_t(Z) + \sum_{\phi(a^J) \leq Z \leq |X|} \mu_t(Z) &\leq \sum_{1 \leq Z \leq \phi(a^J)-1} \mu_s(Z) + \sum_{\phi(a^J) \leq Z \leq |X|} \mu_s(Z) \iff \\ \sum_{1 \leq i \leq J-1} \rho(a^i, A, t) + \frac{k - \phi(a^J)}{|X| - \phi(a^J)} \rho(a^J, A, t) &\leq \sum_{1 \leq i \leq J-1} \rho(a^i, A, s) + \frac{k - \phi(a^J)}{|X| - \phi(a^J)} \rho(a^J, A, s) \\ &\iff \rho(a^J, A, s) \geq \rho(a^J, A, t) \end{aligned}$$

Note that by the construction of $\succ$, the set $\overline{M}(a) = \{b \in X : b \succ a\}$, i.e., the upper contour set of a w.r.t $\succ$. Thus the Axiom 5 is equivalent to for any $a \in A$, $\sum_{\{c \in A : c \succ a\}} \rho(c, A, t) \geq \sum_{\{c \in A : c \succ a\}} \rho(c, A, s)$. Note that by labelling, $a^J \succ c$ for all $c \in A, c \neq a^J$. Let a be such that $a^J \succ a \succ c$ for all $c \in A, c \neq a, a^J$, then the Axiom 5 implies the last line.

If $\phi(a^j) \leq k \leq \phi(a^{j+1})$ for some $1 \leq j \leq J-1$,

$$\begin{aligned} \sum_{1 \leq Z \leq k} \mu_t(Z) &\leq \sum_{1 \leq Z \leq k} \mu_s(Z) \iff \\ \sum_{1 \leq Z \leq \phi(a^j)} \mu_t(Z) + \sum_{\phi(a^j) < Z \leq k} \mu_t(Z) &\leq \sum_{1 \leq Z \leq \phi(a^j)} \mu_s(Z) + \sum_{\phi(a^j) < Z \leq k} \mu_s(Z) \iff \\ \sum_{1 \leq i \leq j-1} \rho(a^i, A, t) + \frac{k - \phi(a^j)}{\phi(a^{j+1}) - \phi(a^j)} \rho(a^j, A, t) &\leq \sum_{1 \leq i \leq j-1} \rho(a^i, A, s) + \frac{k - \phi(a^j)}{\phi(a^{j+1}) - \phi(a^j)} \rho(a^j, A, s) \end{aligned}$$

By the Axiom 5, $\sum_{1 \leq i \leq j-1} \rho(a^i, A, t) \leq \sum_{1 \leq i \leq j-1} \rho(a^i, A, s)$ and $\sum_{1 \leq i \leq j} \rho(a^i, A, t) \leq \sum_{1 \leq i \leq j} \rho(a^i, A, s)$. Suppose the last line above is not true, then it must be the case that

$\rho(a^j, A, t) > \rho(a^j, A, s)$, but this will lead to $\sum_{1 \leq i \leq j} \rho(a^i, A, t) > \sum_{1 \leq i \leq j} \rho(a^i, A, s)$ since $0 \leq \frac{k - \phi(a^j)}{\phi(a^{j+1}) - \phi(a^j)} \leq 1$, which is a contradiction. $\square$

By the Lemma 7, there exists a counting process N that induces $\{\mu_t\}_{t \in \mathcal{T}}$. It can be easily verified that N must satisfy the two properties on the counting process by our construction of μ_t .

Step 4. (Verify ρ is generated by the representation with the constructed $\succ, R, N$.) We next check that ρ has a gradual attention representation with the constructed $(\succ, R, N)$. If $a \in A \setminus \text{supp}(A)$. Since the constructed $\succ, R$ satisfies the support compatibility condition, there exists $b \in A$ such that $b \succ a, bRa$. So for any Z , $A^R(Z)$ either contains both a, b or does not contain a . It follows $\{Z \in \mathbb{Z}^{++} : \mathcal{C}_\succ(A^R(Z)) = a\} = \emptyset$ and therefore $\rho(a, A, t) = 0 = \sum_{\{Z \in \mathbb{Z}^{++} : \mathcal{C}_\succ(A^R(Z)) = a\}} \mu_t(Z)$.

If $a \in \text{supp}(A)$, say $a = a^j$ for some $1 \leq j \leq J$. Then $\{Z \in \mathbb{Z}^{++} : \mathcal{C}_\succ(A^R(Z)) = a\} = \{Z \in \mathbb{Z}^{++} : a^j \in A^R(Z), a^{j+1} \notin A^R(Z)\} = \{\phi(a^j) + 1, \phi(a^j) + 2, \dots, \phi(a^{j+1})\}$ if $j < J$, and is equal to $\{Z \in \mathbb{Z}^{++} : Z \geq \phi(a^J) + 1\}$ if $j = J$. In the former case, $\sum_{\{Z \in \mathbb{Z}^{++} : \mathcal{C}_\succ(A^R(Z)) = a\}} \mu_t(Z) = (\phi(a^{j+1}) - \phi(a^j)) \cdot [\rho(a^j, D, t) / (\phi(a^{j+1}) - \phi(a^j))] = \rho(a^j, D, t)$; and in the latter case, $\sum_{\{Z \in \mathbb{Z}^{++} : \mathcal{C}_\succ(A^R(Z)) = a\}} \mu_t(Z) = 0 + (|X| - \phi(a^J)) \cdot [\rho(a^J, D, t) / (|X| - \phi(a^J))] = \rho(a^J, D, t)$.

Step 5. (Check the uniqueness of $\succ, R$.) Suppose there exists another menu-independent $\succ', R'$ that generates the same ρ . But the Proposition 1 immediately implies $\succ = \succ', R = R'$. $\square$

Proof. (Proof of Proposition 3) It is clear from the Proposition 1 that $b \in \overline{H}(a)$ if and only if bRa , i.e., b is more salient than a . Let $\overline{Z} = \max_{A \in \mathcal{X}, a \in \text{supp}(A)} |\overline{H}(a) \cap A| = |\overline{H}(a^*) \cap A^*|$. Note that such pair of a^*, A^* need not to be unique. Let time $t \in \mathcal{T}$ be arbitrary and we will show how $\mu_t(Z)$ can be arbitrary for $Z > \overline{Z}$ and is identified uniquely for each $1 \leq Z \leq \overline{Z}$.

Note that $\rho(a^*, A^*, t) \leq \sum_{Z > \overline{Z}}^\infty \mu_t(Z)$ because $\overline{Z} = |\overline{H}(a^*) \cap A^*|$ implies there exists $\overline{Z}$ items in A^* that are more salient than a^* . On the other hand, $\max_{A \in \mathcal{X}, a \in \text{supp}(A)} |\overline{H}(a) \cap A| = |\overline{H}(a^*) \cap A^*|$ implies every item in A^* that is less salient than a^* is not never chosen and thus $\rho(a^*, A^*, t) = \sum_{Z > \overline{Z}}^\infty \mu_t(Z)$. We conclude that for any $Z > \overline{Z}$, $\mu_t(Z)$ can be arbitrary with the only constraint $\sum_{Z > \overline{Z}}^\infty \mu_t(Z) = \rho(a^*, A^*, t)$ by the definition of the maximum.

It also follows that $\sum_{a \in \overline{H}(a^*) \cap A^*} \rho(a, A^*, t) = 1 - \sum_{Z > \overline{Z}}^\infty \mu_t(Z) = \sum_{1 \leq Z \leq \overline{Z}} \mu_t(Z)$ since ρ adds up to 1. Since $R \in \mathcal{S}$, the elements in $\overline{H}(a^*) \cap A^*$ can be totally ordered by R and label them by $a_1 R a_2 R \dots R a_{\overline{Z}}$. Now consider the menu $A^* \setminus \{a_{\overline{Z}}\}$. It is clear that $a^* \in \text{supp}(A^* \setminus \{a_{\overline{Z}}\})$ and $\overline{Z} - 1 = |\overline{H}(a^*) \cap A^* \setminus \{a_{\overline{Z}}\}| = \max_{a \in \text{supp}(A^* \setminus \{a_{\overline{Z}}\})} |\overline{H}(a) \cap A^* \setminus \{a_{\overline{Z}}\}|$. For the same reason, we have $\sum_{a \in \overline{H}(a^*) \cap A^* \setminus \{a_{\overline{Z}}\}} \rho(a, A^* \setminus \{a_{\overline{Z}}\}, t) = \sum_{1 \leq Z \leq \overline{Z} - 1} \mu_t(Z)$ and therefore $\mu_t(\overline{Z}) = \sum_{a \in \overline{H}(a^*) \cap A^*} \rho(a, A^*, t) - \sum_{a \in \overline{H}(a^*) \cap A^* \setminus \{a_{\overline{Z}}\}} \rho(a, A^* \setminus \{a_{\overline{Z}}\}, t)$.

We can continue this procedure and for each $Z = 1, \dots, \bar{Z} - 1$, identify each $\mu_t(Z)$ as

$$\sum_{a \in \bar{H}(a^*) \cap A^* \setminus \{a_{\bar{Z}}, a_{\bar{Z}-1}, \dots, a_{Z+1}\}} \rho(a, A^* \setminus \{a_{\bar{Z}}, a_{\bar{Z}-1}, \dots, a_{Z+1}\}, t) - \\ \sum_{a \in \bar{H}(a^*) \cap A^* \setminus \{a_{\bar{Z}}, a_{\bar{Z}-1}, \dots, a_Z\}} \rho(a, A^* \setminus \{a_{\bar{Z}}, a_{\bar{Z}-1}, \dots, a_Z\}, t)$$

The uniqueness of $\mu_t(Z)$ for $1 \leq Z \leq \bar{Z}$ follows immediately by construction. $\square$

Proof. (Proof of Proposition 4) The only if part for the Axiom 6 is easy to check. For the if part, the Theorem 1 shows ρ has a gradual attention representation with menu independent $\succ, R$ and menu dependent μ under the Axiom 1-5. It remains to show how to construct a menu independent μ , given the Axiom 6. Following the proof of Proposition 3, we let $\bar{Z} = \max_{A \in \mathcal{X}, a \in \text{supp}(A)} |\bar{H}(a) \cap A| = |\bar{H}(a^*) \cap A^*|$. Again the choice of a^*, A^* can be arbitrary. For an arbitrary $t \in \mathcal{X}$, we construct μ_t as follows. Label the elements in $\bar{H}(a^*) \cap A^*$ by $a_1 R a_2 R \dots R a_{\bar{Z}}$. Define $\mu_t(\bar{Z}) = \sum_{a \in \bar{H}(a^*) \cap A^*} \rho(a, A^*, t) - \sum_{a \in \bar{H}(a^*) \cap A^* \setminus \{a_{\bar{Z}}\}} \rho(a, A^* \setminus \{a_{\bar{Z}}\}, t)$. For each $Z = 1, \dots, \bar{Z} - 1$, define $\mu_t(Z)$ by

$$\sum_{a \in \bar{H}(a^*) \cap A^* \setminus \{a_{\bar{Z}}, a_{\bar{Z}-1}, \dots, a_{Z+1}\}} \rho(a, A^* \setminus \{a_{\bar{Z}}, a_{\bar{Z}-1}, \dots, a_{Z+1}\}, t) - \\ \sum_{a \in \bar{H}(a^*) \cap A^* \setminus \{a_{\bar{Z}}, a_{\bar{Z}-1}, \dots, a_Z\}} \rho(a, A^* \setminus \{a_{\bar{Z}}, a_{\bar{Z}-1}, \dots, a_Z\}, t)$$

We next show the constructed μ_t is compatible with ρ for arbitrary menus. Let a menu B and $b \in \text{supp}(B)$ be arbitrary. Since ρ has a gradual attention representation with a menu dependent counting process, we denote ν as its induced probability measure over $\mathbb{Z}^+$ for menu B . If $c R b$ for all $c \in \text{supp}(B) \setminus \{b\}$, or equivalently $c \in \text{supp}(B) \setminus \{b\} \implies c \in \bar{H}(b) \cap B$, then $\rho(b, B, t) = 1 - \sum_{1 \leq Z \leq |\bar{H}(b) \cap B|} \nu_t(Z) = 1 - \sum_{c \in \bar{H}(b) \cap B} \rho(c, B, t)$ by the representation. We want to prove $\sum_{1 \leq Z \leq |\bar{H}(b) \cap B|} \mu_t(Z) = \sum_{1 \leq Z \leq |\bar{H}(b) \cap B|} \nu_t(Z)$. Since $|\bar{H}(b) \cap B| = |\bar{H}(a^*) \cap A^* \setminus \{a_{\bar{Z}}, a_{\bar{Z}-1}, \dots, a_{|\bar{H}(b) \cap B|+1}\}|$ and by Axiom 6,

$$\sum_{c \in \bar{H}(b) \cap B} \rho(c, B, t) = \sum_{a \in \bar{H}(a^*) \cap A^* \setminus \{a_{\bar{Z}}, a_{\bar{Z}-1}, \dots, a_{|\bar{H}(b) \cap B|+1}\}} \rho(a, A^* \setminus \{a_{\bar{Z}}, a_{\bar{Z}-1}, \dots, a_{|\bar{H}(b) \cap B|+1}\}, t) \\ = \sum_{1 \leq Z \leq |\bar{H}(b) \cap B|} \mu_t(Z) \text{ by the construction of } \mu_t$$

If there exists $c \in \text{supp}(B)$ and $a R c$ (or equivalently $c \notin \bar{H}(b) \cap B$), then let b^* be the R -maximal element of $\text{supp}(B) \setminus (\bar{H}(b) \cup \{b\})$, or equivalently, $b^* := \arg \max_{\{c: c \in \text{supp}(B), b \in \bar{H}(c)\}} |\bar{H}(c) \cap B|$.

Then $\rho(b, B, t) = \sum_{|\overline{H}(b) \cap B| < Z \leq |\overline{H}(b^*) \cap B|} \nu_t(Z) = \sum_{c \in \overline{H}(b^*) \cap B} \rho(c, B, t) - \sum_{c \in \overline{H}(b) \cap B} \rho(c, B, t)$ by the representation. We want to prove that $\sum_{|\overline{H}(b) \cap B| < Z \leq |\overline{H}(b^*) \cap B|} \nu_t(Z) = \sum_{|\overline{H}(b) \cap B| < Z \leq |\overline{H}(b^*) \cap B|} \mu_t(Z)$. Since

$$\begin{aligned} |\overline{H}(b) \cap B| &= \left| \overline{H}(a^*) \cap A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_{|\overline{H}(b) \cap B|+1}\} \right| \\ |\overline{H}(b^*) \cap B| &= \left| \overline{H}(a^*) \cap A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_{|\overline{H}(b^*) \cap B|+1}\} \right| \end{aligned}$$

, by Axiom 6,

$$\begin{aligned} \sum_{c \in \overline{H}(b^*) \cap B} \rho(c, B, t) - \sum_{c \in \overline{H}(b) \cap B} \rho(c, B, t) &= \\ \sum_{a \in \overline{H}(a^*) \cap A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_{|\overline{H}(b) \cap B|+1}\}} \rho(a, A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_{|\overline{H}(b) \cap B|+1}\}, t) - \\ \sum_{a \in \overline{H}(a^*) \cap A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_{|\overline{H}(b^*) \cap B|+1}\}} \rho(a, A^* \setminus \{a_{\overline{Z}}, a_{\overline{Z}-1}, \dots, a_{|\overline{H}(b^*) \cap B|+1}\}, t) \\ &= \sum_{|\overline{H}(b) \cap B| < Z \leq |\overline{H}(b^*) \cap B|} \mu_t(Z) \text{ by the construction of } \mu_t \end{aligned}$$

□

Proof. (Proof of Corollary 2) It is sufficient to show for any realization of $N_t = Z \in \mathbb{Z}^{++}$, $\mathcal{C}_{\succ}(A^{R_1}(Z)) = \mathcal{C}_{\succ}(A^{R_2}(Z))$ or $\mathcal{C}_{\succ}(A^{R_1}(Z)) \succ \mathcal{C}_{\succ}(A^{R_2}(Z))$. Let $\mathcal{C}_{\succ}(A^{R_2}(Z)) = a$ and suppose $a \succ b$ for any $b \in A^{R_1}(Z)$. If $a R_2 b$, then R_1 being more aligned with $\succ$ implies $a R_1 b$, a contradiction to $a \notin A^{R_1}(Z)$. So we have $b R_2 a$ for any $b \in A^{R_1}(Z)$ but then $A^{R_2}(Z)$ must contain $A^{R_1}(Z)$ and a , a contradiction to $|A^{R_1}(Z)| = |A^{R_2}(Z)|$. □

Proof. (Proof of Proposition 6) 1 $\implies$ 2. Let A be an arbitrary menu. It is equivalent to show that $A \setminus \text{supp}_2(A) \subseteq A \setminus \text{supp}_1(A)$. Let $a \in A \setminus \text{supp}_2(A)$ be arbitrary. Then by the Axiom 4, there exists $b \in A$ such that $b \succ a, b R_2 a$. Since R_1 is more aligned with $\succ$ than R_2 , we have $b R_1 a$ and $a \notin \text{supp}_1(A)$. This concludes $\text{supp}_1(A) \subseteq \text{supp}_2(A)$.

2 $\implies$ 1. Let $a, b \in X$ be arbitrary and say $a \succ b, a R_2 b$. Consider the doubleton menu $\{a, b\}$ and $\text{supp}_2(\{a, b\}) = \{a\}$ since b is dominated by a . Since $\text{supp}_1(\{a, b\}) \subseteq \text{supp}_2(\{a, b\})$ and $\text{supp}_1(\{a, b\})$ is nonempty, we have $\text{supp}_1(\{a, b\}) = \{a\}$, which implies $a \succ b, a R_1 b$. □

Proof. (Proof of Proposition 7) 1 $\implies$ 2. Let A be an arbitrary menu. It is equivalent to show that $A \setminus \text{supp}_2(A) \subseteq A \setminus \text{supp}_1(A)$. Let $a \in A \setminus \text{supp}_2(A)$ be arbitrary. Then by the Axiom 4, there exists $b \in A$ such that $b \succ_2 a, b R a$. Since $\succ_1$ is more aligned with R than $\succ_2$, we have $b \succ_1 a$ and $a \notin \text{supp}_1(A)$. This concludes $\text{supp}_1(A) \subseteq \text{supp}_2(A)$.

2 $\implies$ 1. Let $a, b \in X$ be arbitrary and say $a \succ_2 b, aRb$. Consider the doubleton menu $\{a, b\}$ and $\text{supp}_2(\{a, b\}) = \{a\}$ since b is dominated by a . Since $\text{supp}_1(\{a, b\}) \subseteq \text{supp}_2(\{a, b\})$ and $\text{supp}_1(\{a, b\})$ is nonempty, we have $\text{supp}_1(\{a, b\}) = \{a\}$, which implies $a \succ_1 b, aRb$. $\square$

Proof. (Proof of Proposition 8) The only if part follows immediately from the Proposition 2 with $\succ$ as the desirable P . We prove the if part.

Fix a menu A , let $\text{supp}(A) = \{a \in A : \rho(a, A, t) > 0 \text{ for some } t \in \mathcal{T}\}$, i.e., those items that have been chosen with strictly positive probability at certain time $t \in \mathcal{T}$. Suppose $\rho(\cdot, A, t) \succ_F \rho(\cdot, A, s)$ w.r.t some order $P \in \mathcal{S}$ for all $t > s$. So we can label the items in $\text{supp}(A)$ (say $|\text{supp}(A)| = J$) according to P as $A = \{a^1, a^2, \dots, a^J\}$ such that $a^j P a^i$ if $i < j$. So we have $\sum_{i=1}^j \rho(a^i, A, t) \leq \sum_{i=1}^j \rho(a^i, A, s)$ for all $j = 1, 2, \dots, J$ and for all $t \geq s$. Notice that such labelling is not necessarily unique since P is may not be unique. For example, it is possible that two items b, c are always being chosen with the same strictly positive probability at all time. In such case, we arbitrarily choose one possible labelling.

Define the menu dependent preference $\succ_A$ over $\text{supp}(A)$ by $a^i \succ_A a^j$ if $i > j$. Given $\succ_A$ over $\text{supp}(A)$, we next define the menu-dependent attention order R_A on $\text{supp}(A)$ such that R_A is perfectly misaligned with $\succ_A$ on $\text{supp}(A)$, i.e., $aR_A b \iff b \succ_A a$. We then extend $\succ_A$ to A by $a \succ_A b$ for any $a \in \text{supp}(A), b \in A \setminus \text{supp}(A)$ and extend $\succ_A$ arbitrarily for those $b, c \in A \setminus \text{supp}(A)$. Similarly, We extend R_A to A by $aR_A b$ for any $a \in \text{supp}(A), b \in A \setminus \text{supp}(A)$ and extend R_A arbitrarily for those $b, c \in A \setminus \text{supp}(A)$.

Now consider a probability distribution μ_t over $\mathbb{Z}^{++}$ that is constructed as follows: for any $t \in \mathcal{T}$,

- If $J \geq 2$. For any $Z \in \mathbb{Z}^{++}, 1 \leq Z \leq J-1$, let $\mu_t(Z) = \rho(a^Z, A, t)$; for any $Z \in \mathbb{Z}^{++}, J \leq Z \leq |X|$, let $\mu_t(Z) = (1 - \sum_{1 \leq k \leq J-1} \rho(a^k, A, t)) / (|X| - J + 1)$.
- If $J = 1$. For any $Z \in \mathbb{Z}^{++}, 1 \leq Z \leq |X|$, let $\mu_t(Z) = 1/|X|$

Since $\sum_{i=1}^j \rho(a^i, A, t) \leq \sum_{i=1}^j \rho(a^i, A, s)$ for all $j = 1, 2, \dots, J$ and for all $t \geq s$, it follows $\mu_t \succ_F \mu_s$ w.r.t the natural order $>$ over $\mathbb{Z}^{++}$ for all $t \geq s$. By the Lemma 7, there exists a counting process N^A that induces $\{\mu_t\}_{t \in \mathcal{T}}$. We need to make sure that N^A satisfies the two required assumptions on the counting process but it can be easily verified that they are indeed satisfied by the construction of μ_t .

It remains to check that ρ is indeed generated by the gradual attention model with $(\succ_A, R_A, N^A)$. If $a \in A \setminus \text{supp}(A)$, then there exists $b \in \text{supp}(A)$ such that $b \succ_A a, bR_A a$ and hence for any Z , $A^R(Z)$ either contains both a, b or does not contain a . Thus $\{Z \in \mathbb{Z}^{++} : \mathcal{C}_{\succ_A}(A^R(Z)) = a\} = \emptyset$ and $\rho(a, A, t) = 0 = \sum_{\{Z \in \mathbb{Z}^{++} : \mathcal{C}_{\succ_A}(A^R(Z)) = a\}} \mu_t(Z)$.

If $a \in \text{supp}(A)$, say $a = a^j$ for some $1 \leq j \leq J$. Then $\{Z \in \mathbb{Z}^{++} : \mathcal{C}_{\succ_A}(A^R(Z)) = a\} = \{Z \in \mathbb{Z}^{++} : A^R(Z) = \{a^1, a^2, \dots, a^j\}\} = \{j\}$ if $j < J$, and is equal to $\{Z \in \mathbb{Z}^{++} : Z \geq J\}$ if $j = J$ since a^j is chosen only when it is in the consideration set and any a^i with $i > j$ is not in the consideration set. In either case, $\rho(a, A, t) = \sum_{\{Z \in \mathbb{Z}^{++} : \mathcal{C}_{\succ_A}(A^R(Z)) = a\}} \mu_t(Z)$.

The identification statements on the relationship between $P, \succ_A, R$ follow immediately from the representation. $\square$

Proof. (Proof of Lemma 2) The “if part”. Condition 1 implies deterministic choices and $\rho(a, A) = 1 \iff a \succ b, aRb$ for all $b \in A$. It follows we can uniquely identify $\succ, R$ from all doubleton menus.

Now we consider condition 2. We first claim that positivity $\implies \succ, R$ are perfectly misaligned, i.e., $a \succ b \iff bRa$ for any $a, b \in X$. This is because otherwise there exists two alternatives a, b such that $a \succ b, aRb \implies \rho(b, X) = 0$, a contradiction. Let $a_1 \succ a_2 \succ a_3$ and $a_3Ra_2Ra_1$. Since ρ has a gradual attention representation without time variation, we know $\rho(a, \{a, b\}) \in \{\mu(1), \mu(\geq 2)\}$ and $\rho(a, X) \in \{\mu(1), \mu(2), \mu(\geq 3)\}$. We know $\mu(1) \neq \mu(\geq 2)$ because $\rho(\cdot, \{a, b\}) \neq \frac{1}{2}$ for all $a, b \in X$. Since $\mu(Z) > 0$ for all $Z \in \mathbb{Z}$, $\mu(\geq 3) < \mu(\geq 2)$.

There exists exactly one alternative say b , such that $\rho(b, \{a, b\}) \neq \rho(b, \{b, c\})$. This alternative b corresponds to a_2 . It is unique because $\rho(a_1, \{a_1, a_2\}) = \mu(\geq 2) = \rho(a_1, \{a_1, a_3\})$ and $\rho(a_3, \{a_1, a_3\}) = \mu(1) = \rho(a_3, \{a_2, a_3\})$. Now consider the impact of adding b to the doubleton menu $\{a, c\}$. There will be exactly one alternative say c , such that $\rho(c, \{a, c\}) = \rho(c, \{a, b, c\})$. This alternative c corresponds to a_3 . This is because $\rho(a_1, \{a_1, a_3\}) = \mu(\geq 2) > \mu(\geq 3) = \rho(a_1, \{a_1, a_2, a_3\})$. We label a as a_1 . The identification is then done.

The “only if part”. Suppose all three conditions fail. We want to show there exists two pair of $\succ_1, R_1, \succ_2, R_2$ that can generate the same random choice rule ρ . There are only three possible cases (we leave it to the reader to verify this claim).

Case 1. Positivity holds but $\rho(\cdot, \{a, b\}) = \frac{1}{2}$ for some $a, b \in X$. But since ρ has a gradual attention representation without time variation, it follows $\rho(\cdot, \{b, c\}) = \frac{1}{2}$ for all $b, c \in X$. It also follows that there exists exactly one alternative, say c , such that $\rho(c, X) = \frac{1}{2}$. Consider the following two sets of $\succ_1, R_1, \mu_1, \succ_2, R_2, \mu_2$:

$$\begin{aligned} a \succ_1 b \succ_1 c \quad cR_1bR_1a \quad \mu_1(1) &= \frac{1}{2}, \quad \mu_1(\geq 3) = \rho(b, X) \\ b \succ_2 a \succ_2 c \quad cR_1aR_1b \quad \mu_2(1) &= \frac{1}{2}, \quad \mu_2(\geq 3) = \rho(a, X) \end{aligned}$$

It can be easily verified that the above two sets both generate ρ . (It turns out the above two sets are the only possible two sets, which we will discuss later.)

Case 2. Positivity fails for X and exactly one doubleton menu. Say $\rho(a, \{a, c\}) = 1$. The following two sets of $\succ_1, R_1, \mu_1, \succ_2, R_2, \mu_2$ will work.

$$\begin{aligned} a \succ_1 c \succ_1 b \quad bR_1aR_1c \quad \mu_1(\geq 2) &= \rho(b, X) \\ b \succ_2 a \succ_2 c \quad aR_2cR_2b \quad \mu_2(\geq 2) &= \rho(a, X) \end{aligned}$$

(It turns out the above two sets are the only possible two sets, which we will discuss later.)

Case 3. Positivity fails for X and exactly two doubleton menus. If $\rho(a, X) = 1$, then it must be the case that $\rho(a, \{a, c\}) = \rho(a, \{a, b\}) = 1, \rho(b, \{b, c\}) \in (0, 1)$. The following two sets of $\succ_1, R_1, \mu_1, \succ_2, R_2, \mu_2$ will work.

$$\begin{aligned} a \succ_1 b \succ_1 c \quad aR_1cR_1b \quad \mu_1(\geq 2) &= \rho(b, X) \\ a \succ_2 c \succ_2 b \quad aR_2bR_2c \quad \mu_2(\geq 2) &= \rho(c, X) \end{aligned}$$

If $\text{supp}(\rho(\cdot, X)) = \{a, b\}$, then it must be the case that $\rho(c, \{a, c\}) = \rho(c, \{b, c\}) = 0, \rho(a, \{a, b\}) \in (0, 1)$. The following two sets of $\succ_1, R_1, \mu_1, \succ_2, R_2, \mu_2$ will work.

$$\begin{aligned} a \succ_1 b \succ_1 c \quad bR_1aR_1c \quad \mu_1(\geq 2) &= \rho(a, X) \\ b \succ_2 a \succ_2 c \quad aR_2bR_2c \quad \mu_2(\geq 2) &= \rho(b, X) \end{aligned}$$

□

Proof. (Proof of Lemma 3) We can (vacuously) define $c \sim c$ for all $c \in X$ so $\sim$ is reflexive. Suppose $a \sim b$ ($a \neq b$) and there exists $c \neq a \neq b$ such that $b \sim c$. We want to show such c cannot exist by yielding a contradiction. Let $X' = \{a, b, c\}$ and consider the restriction of ρ on subsets of X' . By the definition of $\sim$ and the Lemma 2, ρ must fall into one of the following three different cases.

Case 1. Positivity holds but $\rho(\cdot, \{a, b\}) = \frac{1}{2}$ for some $a, b \in X$. But since ρ has a gradual attention representation without time variation, it follows $\rho(\cdot, \{b, c\}) = \frac{1}{2}$ for all $b, c \in X'$. It also follows that there exists exactly one item, say a^* , such that $\rho(a^*, X') = \frac{1}{2}$, and $a^*Ra', a' \succ a^*$ for all $a' \in X' \setminus a^*$. But $a \sim b \implies a^* \neq a \neq b$ so the only possible candidate for a^* is c , but $b \sim c$ is then impossible.

Case 2. Positivity fails for X' and exactly one doubleton menu. But this doubleton menu cannot be $\{a, b\}$ because otherwise $a \sim b$ cannot hold. It also cannot be $\{b, c\}$ for the same reason.

Suppose $\rho(a, \{a, c\}) = 1$, then $a \succ c, aRc$. If $b \succ a \succ c$, then positivity of $\{b, c\} \implies aRcRb$ and hence $\succ, R$ is uniquely identified, a contradiction to $a \sim b$. Similarly, if $a \succ b \succ c$,

then positivity of $\{b, c\} \implies aRcRb$ and hence $\succ, R$ is uniquely identified, a contradiction to $a \sim b$. Finally, if $a \succ c \succ b$, then positivity of $\{a, b\} \implies bRaRc$ and hence $\succ, R$ is uniquely identified, a contradiction to $a \sim b$.

Suppose $\rho(c, \{a, c\}) = 1$, then $c \succ a, cRa$. If $b \succ c \succ a$, then positivity of $\{a, b\} \implies cRaRb$ and hence $\succ, R$ is uniquely identified, a contradiction to $a \sim b$. If $c \succ b \succ a$, then positivity of $\{a, b\} \implies cRaRb$ and hence $\succ, R$ is uniquely identified, a contradiction to $a \sim b$. Finally, if $c \succ a \succ b$, then positivity of $\{a, c\} \implies aRcRb$ and hence $\succ, R$ is uniquely identified, a contradiction to $a \sim b$.

Case 3. Positivity fails for X' and exactly two doubleton menu. But $a \sim b \implies$ positivity fails for $\{a, c\}, \{b, c\}$, which is a contradiction to $b \sim c$.

This shows $\sim$ is vacuously transitive, which concludes that $\sim$ is an equivalence relation and each equivalent class can contain at most two elements. $\square$

Proof. (Proof of Theorem 2) It remains to show that the restriction of $\succ, R$ over the equivalent classes are unique. Suppose not, then there exists $\succ_1, R_1$ and $\succ_2, R_2$ such that for some $[a], [c]$, $[a] \succ_1 [c], [c]R_1[a]$ and $[c] \succ_2 [a], [a]R_2[c]$. The other possibilities for $\succ_1, R_1$ and $\succ_2, R_2$ over $[a], [c]$ are ruled out following the same discussion we give after the definition of $\sim$. If both $[a], [c]$ contain only one element, then $a \sim c \implies [a] = [c]$, a contradiction. Otherwise, at least one equivalent class contains two elements, say $[a] = \{a, b\}$, but this implies $a \sim b \sim$, a contradiction $[a]$ containing at most two elements. $\square$

Proof. (Proof of Proposition 9) When $\max_A |supp(A)| \geq 3$, we can obtain full identification if $\rho(\cdot, \{a, b\}) \neq \frac{1}{2}$ for any a, b , by the Lemma 2. It remains to show that the set of primitives that fail this condition has Lebesgue measure zero. Note that $\rho(\cdot, \{a, b\}) \neq \frac{1}{2}$ is equivalent to requiring $\mu(1) \neq \frac{1}{2}$. Consider μ as an element from the unitary simplex in the vector space of $\mathbb{R}^{|X|}$ and the subset of μ s that having $\mu(1)$ clearly has a Lebesgue measure zero. $\square$

Proof. (Proof of Lemma 4) The probability that the support fails this requirement can be bounded above by

$$\frac{3^K \cdot 2^K \cdot K!}{(K!)^2} \rightarrow 0 \quad \text{as} \quad K \rightarrow \infty$$

The upper bound can be thought as assigning K people into K houses, and each house contains two beds. The people, houses, and the beds are indexed. Each house having at most two beds is like we allow $\succ, R$ to be perfectly misaligned over at most two items. The number of houses end up having people must be less than or equal to K . Note that $\succ, R$ restricted on the houses must be perfectly aligned. $\square$

Proof. (Proof of Lemma 5) It is sufficient to show that for any $t > s$, $\bar{\mu}_t \succ_F \bar{\mu}_s$ w.r.t the natural order over $\mathbb{Z}^{++}$ since then by the Lemma 7, there exists a counting process $\bar{N}$ that induces $\bar{\mu}_t$.

Note that for any $K \in \mathbb{Z}^{++}$,

$$\begin{aligned} \sum_{Z=1}^K \bar{\mu}_t(Z) &= \sum_{Z=1}^K \int \mu_t^N(Z) dm^C = \int \left(\sum_{Z=1}^K \mu_t^N(Z) \right) dm^C \\ &\leq \int \left(\sum_{Z=1}^K \mu_s^N(Z) \right) dm^C = \sum_{Z=1}^K \int \mu_s^N(Z) dm^C = \sum_{Z=1}^K \bar{\mu}_s(Z) \end{aligned}$$

The inequality comes from the fact that each $N \in \text{supp}(m^C)$ is a counting process. It can then easily be verified that $\bar{N} \in \mathcal{N}$, i.e., satisfying the restrictions we have imposed on the counting process. $\square$

Proof. (Proof of Proposition 11) We shall first prove whenever a representative agent exists, $R \succ_{\succ} \bar{R}$ for any element $R \in \text{supp}(m^S)$. Note that an item a is not in the support of a menu containing it if and only if there exists an item b such that $b \succ a, bRa$ for all $R \in \text{supp}(m^S)$ if and only if $b \succ a, b\bar{R}a$. Thus for an arbitrary menu, the support under $\bar{R}$ is a subset of the support under any $R \in \text{supp}(m^S)$. The claim hence follows from the Proposition 6.

The if part can then be verified by letting $\bar{R}$ be the minimal element of $\text{supp}(m^S)$ when it can be totally ordered by $\succ_{\succ}$.

The only if part can be proved by the Example 1 which we used for demonstrating non-existence of a representative agent. $\square$

Proof. (Proof of Corollary 3) By the Proposition 8, it is sufficient to prove that $\rho(\cdot, A, t) \succ_F \rho(\cdot, A, s)$ w.r.t $\succ$ for any $t \geq s$. This is because for any $a \in A$,

$$\begin{aligned} \sum_{\{b \in A: a \succ b\}} \rho(b, A, t) &\leq \sum_{\{b \in A: a \succ b\}} \rho(b, A, s) \iff \\ \sum_{\{b \in A: a \succ b\}} \int \rho_R(b, A, t) dm^S(R) &\leq \sum_{\{b \in A: a \succ b\}} \int \rho_R(b, A, s) dm^S(R) \iff \\ \int \left(\sum_{\{b \in A: a \succ b\}} \rho_R(b, A, t) \right) dm^S(R) &\leq \int \left(\sum_{\{b \in A: a \succ b\}} \rho_R(b, A, s) \right) dm^S(R) \end{aligned}$$

And the last line follows from $\sum_{\{b \in A: a \succ b\}} \rho_R(b, A, t) \leq \sum_{\{b \in A: a \succ b\}} \rho_R(b, A, s)$ for any $R \in \mathcal{S}$, by the Proposition 2. $\square$

Proof. (Proof of Corollary 4) Note that the support of the aggregated ρ is $\text{supp}(A) = \cup_{\succ_i \in \text{supp}(m^P)} \text{supp}_i(A)$. We construct $\succ \in \mathcal{S}$ as follows:

- $\succ$ is perfectly misaligned with R over $\text{supp}(A)$ and denote a^* as the most salient item in $\text{supp}(A)$.
- $\succ$ is perfectly aligned with R over $\text{supp}(A)$ and denote b^* as the most salient item in $A \setminus \text{supp}(A)$.
- Let $a^* \succ b^*$.

We claim the constructed $\succ$ agrees with $\succ_i$ over $\text{supp}_i(A)$ for any $\succ_i \in \text{supp}(m^P)$. To see this let $a \succ_i b, a, b \in \text{supp}_i(A) \subseteq \text{supp}(A)$ so bRa since both items are in the support. Suppose $b \succ a$, then $a \notin \text{supp}(A)$, a contradiction. For each $\succ_i \in \text{supp}(m^P)$, we know $\rho_i(\cdot, A, t) \succ_F \rho_i(\cdot, A, s)$ w.r.t $\succ_i$ for any $t \geq s$. It follows from the claim that $\rho_i(\cdot, A, t) \succ_F \rho_i(\cdot, A, s)$ also w.r.t $\succ$ for any $t \geq s$. That is for any $a \in A$,

$$\begin{aligned} \sum_{\{b \in A: a \succ b\}} \rho_i(b, A, t) &\leq \sum_{\{b \in A: a \succ b\}} \rho_i(b, A, s) \text{ for all } \succ_i \in \text{supp}(m^P) \implies \\ \int \left(\sum_{\{b \in A: a \succ b\}} \rho_i(b, A, t) \right) dm^P(\succ_i) &\leq \int \left(\sum_{\{b \in A: a \succ b\}} \rho_i(b, A, s) \right) dm^P(\succ_i) \iff \\ \sum_{\{b \in A: a \succ b\}} \int \rho_i(b, A, t) dm^P(\succ_i) &\leq \sum_{\{b \in A: a \succ b\}} \int \rho_i(b, A, s) dm^P(\succ_i) \iff \\ \sum_{\{b \in A: a \succ b\}} \rho(b, A, t) &\leq \sum_{\{b \in A: a \succ b\}} \rho(b, A, s) \end{aligned}$$

Thus by the Proposition 8, a representative agent exists. $\square$

Proof. (Proof of Proposition 12) We shall first prove whenever a representative agent exists, $\succ_i \succ_R \succ$ for any element $\succ_i \in \text{supp}(m^P)$. Note that an item a is not in the support of a menu containing it if and only if there exists an item b such that $b \succ_i a, bRa$ for all $\succ_i \in \text{supp}(m^P)$ if and only if $b \succ a, bRa$. Thus for an arbitrary menu, the support under $\succ$ is a subset of the support under any $\succ_i \in \text{supp}(m^P)$. The claim hence follows from the Proposition 7.

The if part can then be verified by constructing $\succ$ in the same way as the Corollary 4.

The only if part can be proved by replacing $\succ, R_1, R_2$ with $R, \succ_1, \succ_2$ in the Example 1. $\square$

7.2 Counting Process and FOSD

A counting process $\{N_t\}_{t \in \mathcal{T}}$ is a non-decreasing stochastic process, i.e., for $t \geq s$, $Pr(N_t < N_s | N_s) = 0$, that takes values in $\mathbb{Z}^+$. For any time t , we denote μ_t as its induced probability distribution over $\mathbb{Z}^+$, i.e., $\mu_t(Z) := Pr(N_t = Z)$ (we ignore the superscript N in the appendix). The following lemma follows immediately from the definition of a counting process.

Lemma 6. (*Counting Process implies FOSD*)

Suppose the collection of $\{\mu_t\}_{t \in \mathcal{T}}$ is induced by a counting process $\{N_t\}_{t \in \mathcal{T}}$, then $\mu_t \succ_F \mu_s$ w.r.t the natural order $>$ over $\mathbb{Z}^+$ for all $t \geq s$.

Proof. Let $t \geq s$. For any $K \in \mathbb{Z}^+$,

$$\begin{aligned} \sum_{j=0}^K \mu_t(j) &= \sum_{j=0}^K \sum_{i=0}^j [Pr(N_t = j | N_s = i) \cdot Pr(N_s = i)] \text{ by non-decreasing of } N_t \\ &= \sum_{i=0}^K Pr(N_s = i) \cdot \left(\sum_{j=i}^K Pr(N_t = j | N_s = i) \right) \\ &\leq \sum_{i=0}^K Pr(N_s = i) = \sum_{i=0}^K \mu_s(i) \end{aligned}$$

□

It turns out the converse statement of the Lemma 6 is also true, though there could be multiple counting processes as shown in the Example 2.

Lemma 7. (*FOSD Characterization of the Counting Process*)

Let $\{\mu_t\}_{t \in \mathcal{T}}$ be a collection of probability distributions over $\mathbb{Z}^+$. Suppose $\mu_t \succ_F \mu_s$ w.r.t the natural order $>$ on $\mathbb{Z}^+$ for all $t \geq s$, then there exists a counting process $\{N_t\}_{t \in \mathcal{T}}$ such that it induces μ_t for all t .

Proof. It suffices to show that for any $t \geq s$, there exists random variables N_t, N_s, N_{t-s} distributed over $\mathbb{Z}^+$ such that $Pr(N_t = Z) = \mu_t(Z), Pr(N_s = Z) = \mu_s(Z)$ for all $Z \in \mathbb{Z}^+$ and $N_t = N_s + N_{t-s}$. This is because we can then construct the transition probabilities from N_s to N_t by $Pr(N_t = Z_1 | N_s = Z_2) = Pr(N_{t-s} = Z_1 - Z_2)$ for all $Z_1, Z_2 \in \mathbb{Z}^+$ and $Pr(N_{t-s} < 0) = 0$ implies $Pr(N_t < N_s | N_s) = 0$.

We use F_t, F_s to denote the CDFs for N_t, N_s (over $\mathbb{R}$) and note that $\mu_t \succ_F \mu_s$ implies $F_t(Z) := \sum_{j=0}^Z \mu_t(j) \leq F_s(Z) := \sum_{j=0}^Z \mu_s(j)$ for all $Z \in \mathbb{Z}^+$. Let W be a random variable

that has a uniform distribution over $[0, 1]$. Define a function $G_t : [0, 1] \rightarrow \mathbb{Z}^+$ as follows:

$$G_t(w) := \inf\{Z \in \mathbb{Z}^+ : F_t(Z) \geq w\}$$

Since F_t is a right-continuous step function that jumps only on $\mathbb{Z}^+$, $F_t \circ G_t(w) \geq w$ and $G_t \circ F_t(Z) = Z$ for all $w \in [0, 1], Z \in \mathbb{Z}^+$. We next verify that the random variable $G_t(W)$ has the same CDF as F_t . On one hand, the event $\{G_t(W) \leq Z\} \subseteq \{F_t \circ G_t(W) \leq F_t(Z)\}$ by non-decreasing of F_t so $Pr(G_t(W) \leq Z) \leq Pr(F_t \circ G_t(W) \leq F_t(Z)) \leq Pr(W \leq F_t(Z)) = F_t(Z)$. On the other hand, the event $\{W \leq F_t(Z)\} \subseteq \{G_t(W) \leq G_t \circ F_t(Z)\}$ by non-decreasing of G_t so $F_t(Z) = Pr(W \leq F_t(Z)) \leq Pr(G_t(W) \leq G_t \circ F_t(Z)) = Pr(G_t(W) \leq Z)$. Thus we have $Pr(G_t(W) \leq Z) = F_t(Z)$. We similarly define a function G_s from F_s so $G_s(W)$ has the same CDF as F_s . Let $N_t = G_t(W), N_s = G_s(W)$ and we construct the random variable $N_{t-s} := G_t(W) - G_s(W)$. Since $F_t \leq F_s$ everywhere, $G_t \geq G_s$ everywhere and hence $Pr(N_{t-s} \geq 0) = 1$ as desired. $\square$

We provide an example below to show that the uniqueness result fails in general, i.e., there exists two different counting process that can induce the same collection of distributions. Moreover, the process that induces the distributions does not need to be a counting process.

Example 2. (Multiple Counting Processes)

We give a simple example where the support of $\mu_s, \mu_t (t \geq s)$ contains only $\{1, 2, 3\}$. So we can write μ_s, μ_t as the following vectors:

$$\mu_s = (0.3, 0.3, 0.4) \quad \mu_t = (0.1, 0.2, 0.7)$$

Indeed $\mu_t \succ_F \mu_s$ w.r.t the natural order on $\mathbb{Z}$. Consider the following three transition matrices from s to t . It can be checked that $\mu_s M_i = \mu_t$ for all $i = 1, 2, 3$.

$$M_1 = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad M_2 = \begin{bmatrix} \frac{1}{3} & 0 & \frac{2}{3} \\ 0 & \frac{2}{3} & \frac{1}{3} \\ 0 & 0 & 1 \end{bmatrix} \quad M_3 = \begin{bmatrix} 0 & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ 0 & \frac{1}{4} & \frac{3}{4} \end{bmatrix}$$

It is clear that M_1, M_2 represents two different counting processes while M_3 is not a counting process since there is strict positive probability of decreasing.